CHAPTER 24
Amines and Heterocycles



FIGURE 24.1 **The characteristic and unmistakable odor of fish is due to a mixture of simple alkylamines**. (credit: modification of work "Pike Place" by Sharat Ganapati/Flickr, CC BY 2.0)

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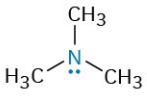
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WHY THIS CHAPTER? By the end of this chapter, we will have seen all the common functional groups. Of those groups, carbonyl compounds and amines are the most abundant and have the richest chemistry. In addition to proteins and nucleic acids, the majority of pharmaceutical agents contain amine functional groups, and many of the common coenzymes necessary for biological catalysis are amines.

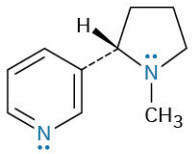
Amines are organic derivatives of ammonia in the same way that alcohols and ethers are organic derivatives of water. Like ammonia, amines contain a nitrogen atom with a lone pair of electrons, making amines both basic and nucleophilic. We'll soon see, in fact, that most of the chemistry of amines depends on the presence of this lone pair of electrons.

Amines occur widely in all living organisms. Trimethylamine, for instance, occurs in animal tissues and is partially responsible for the distinctive odor of fish; nicotine is found in tobacco; and cocaine is a stimulant found in the leaves of the South American coca bush. In addition, amino acids are the building blocks from

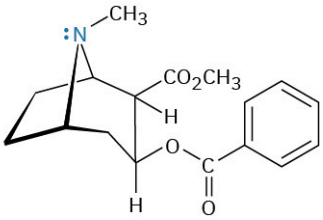
which all proteins are made, and cyclic amine bases are constituents of nucleic acids.



Trimethylamine



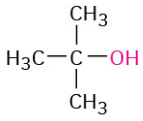
Nicotine



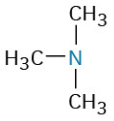
Cocaine

24.1 Naming Amines

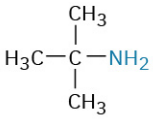
Amines can be either alkyl-substituted (**alkylamines**) or aryl-substituted (**arylamines**). Although much of the chemistry of the two classes is similar, there are also substantial differences. Amines are classified as **primary (RNH_2)**, **secondary (R_2NH)**, or **tertiary (R_3N)**, depending on the number of organic substituents attached to nitrogen. Thus, methylamine (CH_3NH_2) is a primary amine, dimethylamine [$(CH_3)_2NH$] is a secondary amine, and trimethylamine [$(CH_3)_3N$] is a tertiary amine. Note that this usage of the terms *primary, secondary,* and *tertiary* differs from our previous usage. When we speak of a tertiary alcohol or alkyl halide, we refer to the degree of substitution at the alkyl carbon atom, but when we speak of a tertiary amine, we refer to the degree of substitution at the nitrogen atom.



***tert*-Butyl alcohol (a tertiary alcohol)**

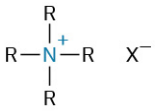


Trimethylamine (a tertiary amine)



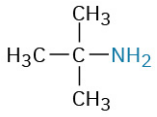
***tert*-Butylamine (a primary amine)**

Compounds containing a nitrogen atom with four attached groups also exist, but the nitrogen atom must carry a formal positive charge. Such compounds are called **quaternary ammonium salts**.

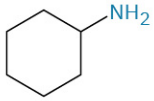


A quaternary ammonium salt

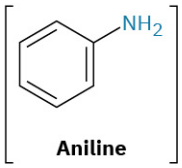
Primary amines are named in the IUPAC system in several ways. For simple amines, the suffix *-amine* is added to the name of the alkyl substituent. You might also recall from the chapter on **Benzene and Aromaticity** that the aromatic phenylamine, $H_2N–C_6H_5$, has the common name *aniline.*



***tert*-Butylamine**

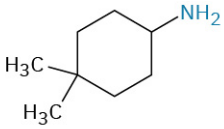


Cyclohexylamine

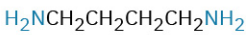


Aniline

Alternatively, the suffix *-amine* can be used in place of the final *-e* in the name of the parent compound.

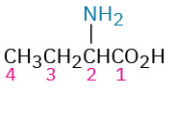


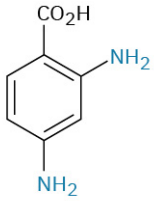
4,4-Dimethylcyclohexanamine

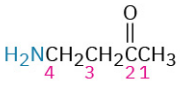


1,4-Butanediamine

Amines with more than one functional group are named by considering the $–NH_2$ as an *amino* substituent on the parent molecule.

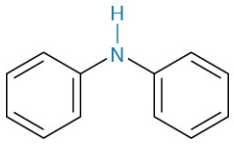


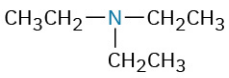




2-Aminobutanoic acid **2,4-Diamino**benzoic acid **4-Amino-2-**butanone

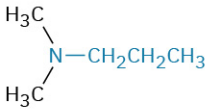
Symmetrical secondary and tertiary amines are named by adding the prefix *di-* or *tri-* to the alkyl group.

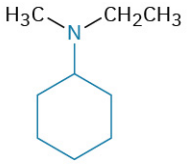




Diphenylamine **Triethylamine**

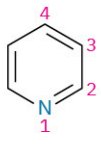
Unsymmetrically substituted secondary and tertiary amines are referred to as *N*-substituted primary amines. The largest alkyl group takes the parent name, and the other alkyl groups are considered *N*-substituents on the parent (*N* because they're attached to nitrogen).

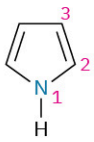


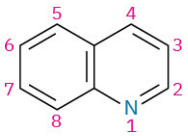


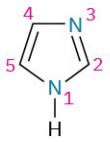
***N,N*-Dimethylpropylamine** ***N*-Ethyl-*N*-methylcyclohexylamine**

Heterocyclic amines—compounds in which the nitrogen atom occurs as part of a ring—are also common, and each different heterocyclic ring system has its own parent name. The heterocyclic nitrogen atom is always numbered as position 1.

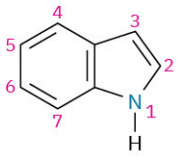


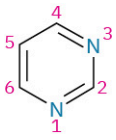


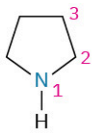


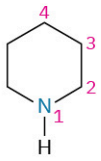


Pyridine **Pyrrole** **Quinoline** **Imidazole**





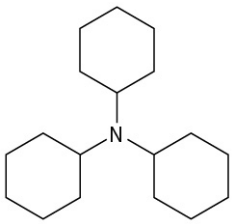




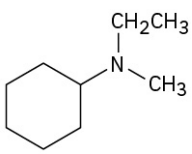
Indole **Pyrimidine** **Pyrrolidine** **Piperidine**

PROBLEM 24-1 Name the following compounds:

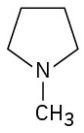
(a) $CH_3NHCH_2CH_3$ **(b)**



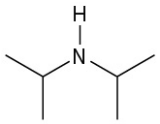
(c)



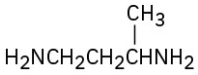
(d)



(e)



(f)



PROBLEM 24-2 Draw structures corresponding to the following IUPAC names:
(a) Triisopropylamine **(b)** Triallylamine **(c)** *N*-Methylaniline
(d) *N*-Ethyl-*N*-methylcyclopentylamine **(e)** *N*-Isopropylcyclohexylamine **(f)** *N*-Ethylpyrrole

PROBLEM 24-3 Draw structures for the following heterocyclic amines:
(a) 5-Methoxyindole **(b)** 1,3-Dimethylpyrrole **(c)** 4-(*N*,*N*-Dimethylamino)pyridine
(d) 5-aminopyrimidine

24.2 Structure and Properties of Amines

The bonding in alkylamines is similar to the bonding in ammonia. The nitrogen atom is sp^3-hybridized, with its three substituents occupying three corners of a regular tetrahedron and the lone pair of electrons occupying the fourth corner. As you might expect, the C–N–C bond angles are close to the 109° tetrahedral value. For trimethylamine, the C–N–C bond angle is 108° and the C–N bond length is 147 pm.



Trimethylamine

One consequence of tetrahedral geometry is that an amine with three different substituents on nitrogen is chiral, as we saw in **Section 5.10**. Unlike chiral carbon compounds, however, chiral amines can't usually be resolved because the two enantiomeric forms rapidly interconvert by a *pyramidal inversion*, much as an alkyl halide inverts in an S_N2 reaction. Pyramidal inversion occurs by a momentary rehybridization of the nitrogen atom to planar, sp^2 geometry, followed by rehybridization of the planar intermediate to tetrahedral, sp^3 geometry (**FIGURE 24.2**). The barrier to inversion is about 25 kJ/mol (6 kcal/mol), an amount only twice as large as the barrier to rotation about a C–C single bond.

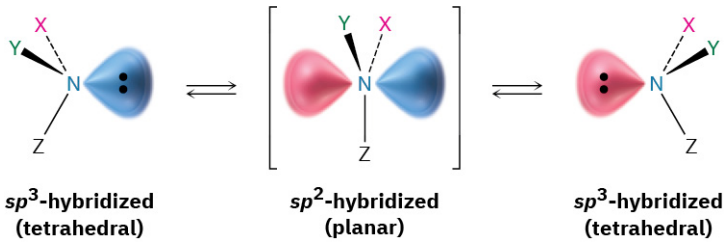
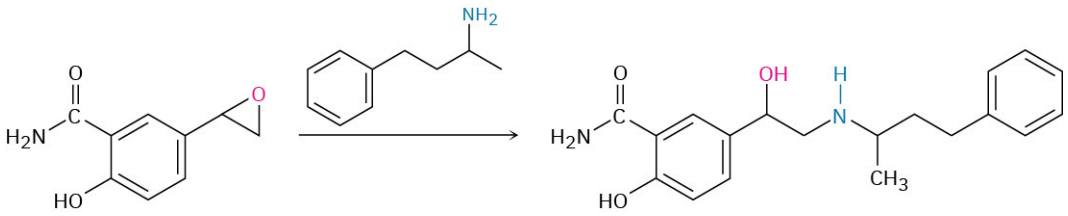


FIGURE 24.2 Pyramidal inversion rapidly interconverts the two mirror-image (enantiomeric) forms of an amine.

Alkylamines have a variety of applications in the chemical industry as starting materials for the preparation of insecticides and pharmaceuticals. Labetalol, for instance, a so-called β-blocker used for the treatment of high blood pressure, is prepared by S_N2 reaction of an epoxide with a primary amine. The substance marketed for drug use is a mixture of all four possible stereoisomers, but the biological activity results primarily from the (*R*,*R*) isomer.



Labetalol

Like alcohols, amines with fewer than five carbon atoms are generally water-soluble. Also like alcohols, primary and secondary amines form hydrogen bonds and are highly associated. As a result, amines have higher boiling points than alkanes of similar molecular weight. Diethylamine (MW = 73 amu) boils at 56.3 °C, for instance, while pentane (MW = 72 amu) boils at 36.1 °C.

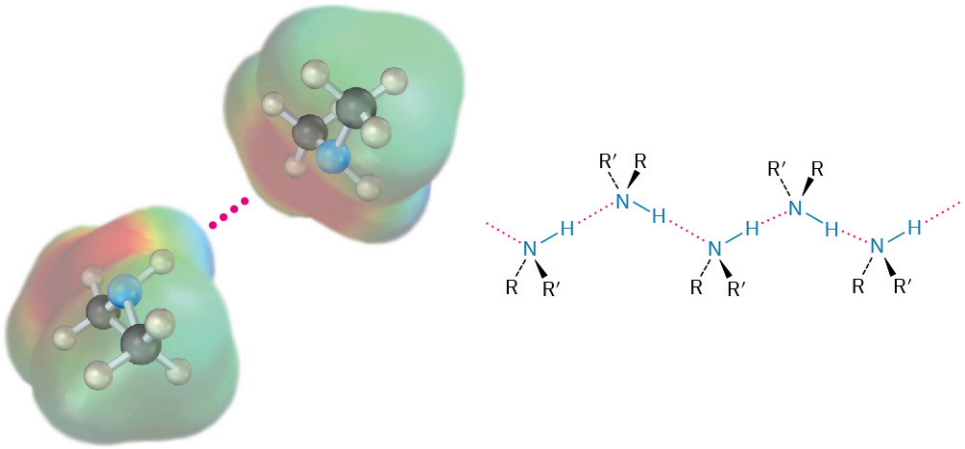
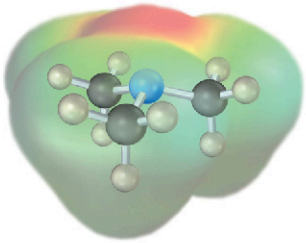


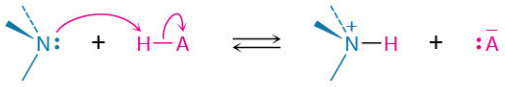
FIGURE 24.3 **Hydrogen bonding in amines.**

One other characteristic of amines is their odor. Low-molecular-weight amines such as trimethylamine have a distinctive fishlike aroma, while diamines such as cadaverine (1,5-pentanediamine) and putrescine (1,4-butanediamine) have the appalling odors you might expect from their common names. Both of these diamines arise from the decomposition of decaying tissue.

24.3 Basicity of Amines

The chemistry of amines is dominated by the lone pair of electrons on nitrogen, which makes amines both basic and nucleophilic. They react with acids to form acid–base salts, and they react with electrophiles in many of the polar reactions seen in past chapters. Note in the following electrostatic potential map of trimethylamine how the negative (red) region corresponds to the lone pair of electrons on nitrogen.





An amine (a Lewis base) **An acid** **A salt**

Amines are much stronger bases than alcohols and ethers, their oxygen-containing analogs. When an amine is dissolved in water, an equilibrium is established in which water acts as an acid and transfers a proton to the amine. Just as the acid strength of a carboxylic acid can be measured by defining an acidity constant K_a (Section 2.8), the base strength of an amine can be measured by defining an analogous *basicity constant* K_b. The larger the value of K_b and the smaller the value of pK_b, the more favorable the proton-transfer equilibrium and the stronger the base.

For the reaction

$$RNH_2 + H_2O \rightleftarrows RNH_3^+ + OH^-$$

$$K_b = \frac{[RNH_3^+]\,[OH^-]}{[RNH_2]}$$

$$pK_b = -\log K_b$$

In practice, K_b values are not often used. Instead, the most convenient way to measure the *basicity* of an amine (RNH_2) is to look at the *acidity* of the corresponding ammonium ion (RNH_3^+).

For the reaction

$$RNH_3^+ + H_2O \rightleftarrows RNH_2 + H_3O^+$$

$$K_a = \frac{[RNH_2]\,[H_3O^+]}{[RNH_3^+]}$$

so

$$K_a \cdot K_b = \left[\frac{[RNH_2]\,[H_3O^+]}{[RNH_3^+]}\right]\left[\frac{[RNH_3^+]\,[OH^-]}{[RNH_2]}\right]$$

$$= [H_3O^+]\,[OH^-] = K_w = 1.00 \times 10^{-14}$$

Thus

$$K_a = \frac{K_w}{K_b} \quad \text{and} \quad K_b = \frac{K_w}{K_a}$$

and

$$pK_a + pK_b = 14$$

These equations say that the K_b of an amine multiplied by the K_a of the corresponding ammonium ion is equal to K_w, the ion-product constant for water (1.00×10^{-14}). Thus, if we know K_a for an ammonium ion, we also know K_b for the corresponding amine base because $K_b = K_w/K_a$. The more acidic the ammonium ion, the less tightly the proton is held and the weaker the corresponding base. That is, a weaker base has an ammonium ion with a smaller pK_a and a stronger base has an ammonium ion with a larger pK_a.

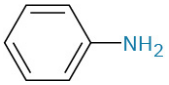
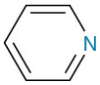
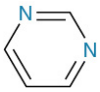
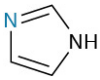
Weaker base	Smaller pK_a for ammonium ion
Stronger base	Larger pK_a for ammonium ion

TABLE 24.1 lists pK_a values of the ammonium ions from a variety of amines and indicates that there is a substantial range of amine basicities. Most simple alkylamines are similar in their base strength, with pK_a's for their ammonium ions in the narrow range 10 to 11. Arylamines, however, are considerably less basic than alkylamines, as are the heterocyclic amines pyridine and pyrrole.

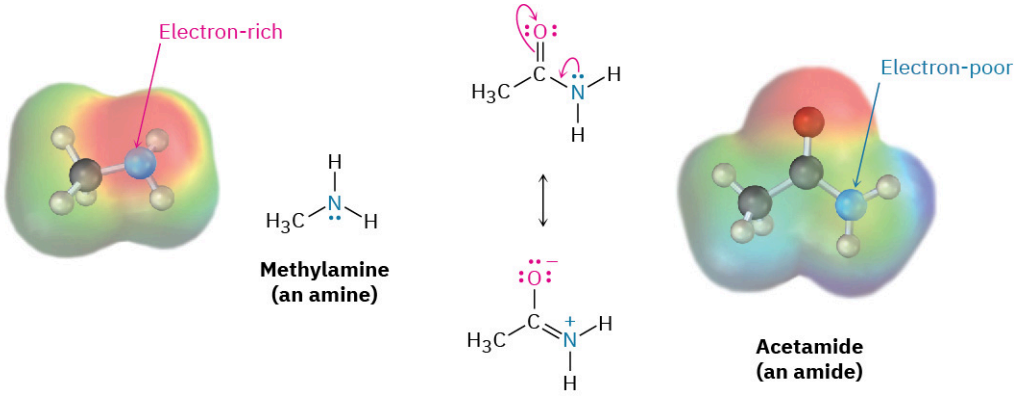
TABLE 24.1 Basicity of Some Common Amines

Name	Structure	pK_a of ammonium ion
Ammonia	NH_3	9.26
Primary alkylamine		
Methylamine	CH_3NH_2	10.64
Ethylamine	$CH_3CH_2NH_2$	10.75
Secondary alkylamine		

TABLE 24.1 Basicity of Some Common Amines

Name	Structure	pK_a of ammonium ion
Diethylamine	$(CH_3CH_2)_2NH$	10.98
Pyrrolidine		11.27
Tertiary alkylamine		
Triethylamine	$(CH_3CH_2)_3N$	10.76
Arylamine		
Aniline		4.63
Heterocyclic amine		
Pyridine		5.25
Pyrimidine		1.3
Pyrrole		0.4
Imidazole		6.95

In contrast with amines, amides ($RCONH_2$) are nonbasic. Amides aren't protonated by aqueous acids, and they are poor nucleophiles. The main reason for this difference in basicity between amines and amides is that an amide is stabilized by delocalization of the nitrogen lone-pair electrons through orbital overlap with the carbonyl group. In resonance terms, amides are more stable and less reactive than amines because they are hybrids of two resonance forms. This amide resonance stabilization is lost when the nitrogen atom is protonated, so protonation is disfavored. The following electrostatic potential maps clearly show a reduced electron density on the amide nitrogen.



To purify amines, it's often possible to take advantage of their basicity. For example, if a mixture of a basic amine and a neutral compound such as a ketone or alcohol is dissolved in an organic solvent and aqueous acid is added, the basic amine dissolves in the water layer as its protonated salt, while the neutral compound remains in the organic solvent layer. Separation of the water layer and neutralization of the ammonium ion by addition of NaOH then provides the pure amine (**FIGURE 24.4**).

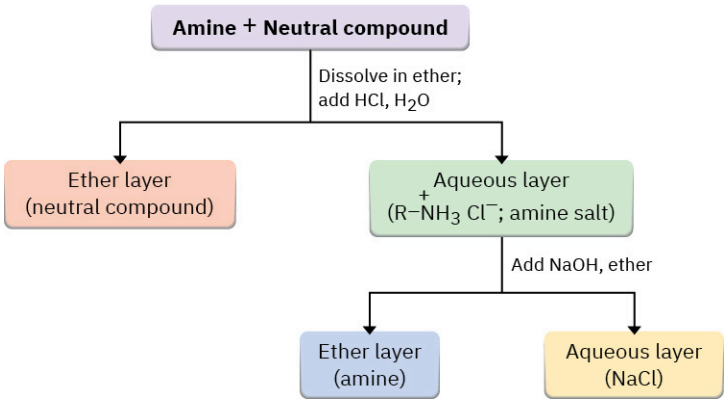
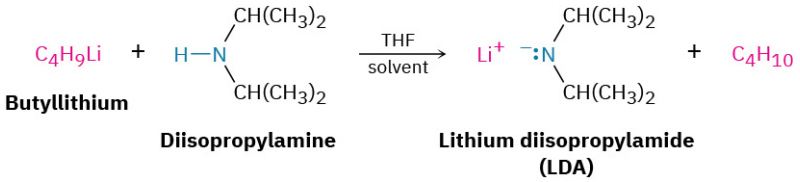


FIGURE 24.4 Separation and purification of an amine component from a mixture by extraction of its ammonium salt into water.

In addition to their behavior as bases, primary and secondary amines can also act as very weak acids because an N–H proton can be removed by a sufficiently strong base. We've seen, for example, how diisopropylamine ($pK_a \approx 36$) reacts with butyllithium to yield lithium diisopropylamide (LDA; **Section 22.5**). Dialkylamine anions like LDA are very strong bases that are often used in laboratory organic chemistry for the generation of enolate ions from carbonyl compounds (**Section 22.7**). They are not, however, encountered in biological chemistry.



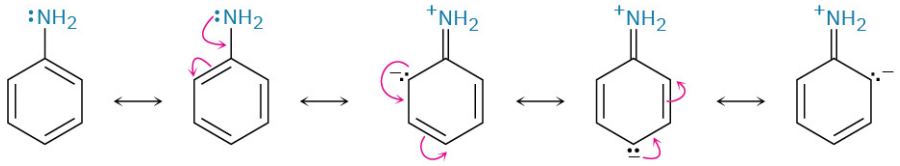
PROBLEM 24-4 Which compound in each of the following pairs is more basic?
(a) $CH_3CH_2NH_2$ or $CH_3CH_2CONH_2$ **(b)** NaOH or CH_3NH_2 **(c)** CH_3NHCH_3 or pyridine

PROBLEM 24-5 The benzylammonium ion ($C_6H_5CH_2NH_3^+$) has $pK_a = 9.33$, and the propylammonium ion has $pK_a = 10.71$. Which is the stronger base, benzylamine or propylamine? What are the pK_b's of benzylamine and propylamine?

24.4 Basicity of Arylamines

As noted previously, arylamines are generally less basic than alkylamines. Anilinium ion has $pK_a = 4.63$, for instance, whereas methylammonium ion has $pK_a = 10.64$. Arylamines are less basic than alkylamines because the nitrogen lone-pair electrons are delocalized by interaction with the aromatic ring's π electron system and are less available for bonding to H^+. In resonance terms, arylamines are stabilized relative to alkylamines

because of their five resonance forms.



Much of the resonance stabilization is lost on protonation, however, so the energy difference between protonated and nonprotonated forms is higher for arylamines than it is for alkylamines. As a result, arylamines are less basic. FIGURE 24.5 illustrates this difference.

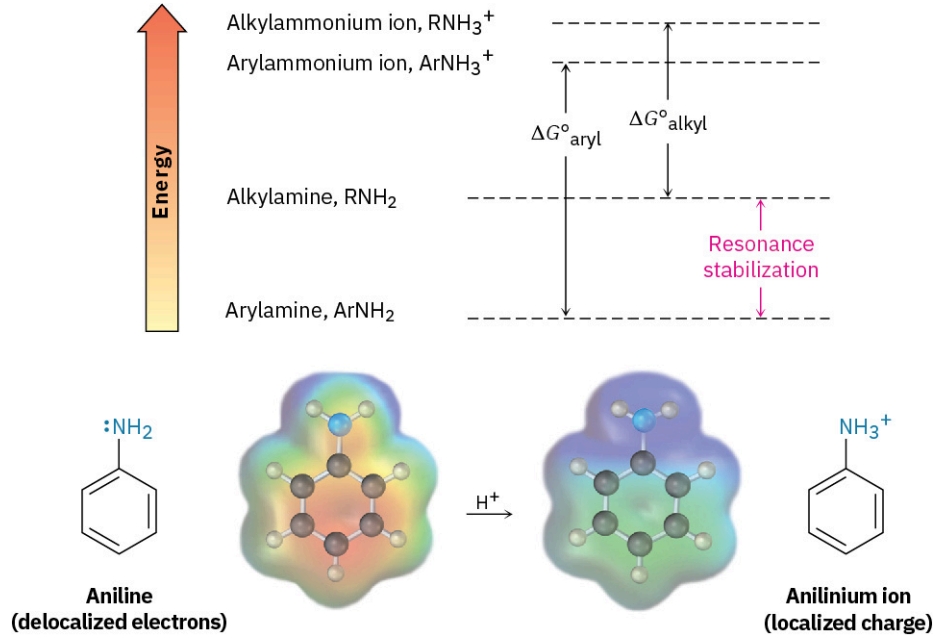


FIGURE 24.5 Arylamines have a larger positive $\Delta G°$ for protonation and are therefore less basic than alkylamines, primarily because of resonance stabilization of the ground state. Electrostatic potential maps show that lone-pair electron density is delocalized in the amine but the charge is localized in the corresponding ammonium ion.

Substituted arylamines can be either more basic or less basic than aniline, depending on the substituent. Electron-donating substituents, such as $-CH_3$, $-NH_2$, and $-OCH_3$, which increase the reactivity of an aromatic ring toward electrophilic substitution (Section 16.4), also increase the basicity of the corresponding arylamine. Electron-withdrawing substituents, such as $-Cl$, $-NO_2$, and $-CN$, which decrease ring reactivity toward electrophilic substitution, also decrease arylamine basicity. TABLE 24.2 considers only *p*-substituted anilines, but similar trends are observed for ortho and meta derivatives.

TABLE 24.2 Base Strength of Some *p*-Substituted Anilines



	Substituent, Y	pK_a	
Stronger base	–NH_2	6.15	Activating groups
	–OCH_3	5.34	
	–CH_3	5.08	
	–H	4.63	
	–Cl	3.98	Deactivating groups
	–Br	3.86	
Weaker base	–CN	1.74	
	–NO_2	1.00	

PROBLEM 24-6 Without looking at Table 24.2, rank the following compounds in order of ascending basicity.

(a) *p*-Nitroaniline, *p*-aminobenzaldehyde, *p*-bromoaniline

(b) *p*-Chloroaniline, *p*-aminoacetophenone, *p*-methylaniline

(c) *p*-(Trifluoromethyl)aniline, *p*-methylaniline, *p*-(fluoromethyl)aniline

24.5 Biological Amines and the Henderson–Hasselbalch Equation

We saw in **Section 20.3** that the extent of dissociation of a carboxylic acid HA in an aqueous solution buffered to a given pH can be calculated with the Henderson–Hasselbalch equation. Furthermore, we concluded that at the physiological pH of 7.3 inside living cells, carboxylic acids are almost entirely dissociated into their carboxylate anions, RCO_2^-.

Henderson–Hasselbalch equation:

$$\text{pH} = \text{p}K\text{a} + \log\frac{[\text{A}^-]}{[\text{HA}]} \quad \text{so} \quad \log\frac{[\text{A}^-]}{[\text{HA}]} = \text{pH} - \text{p}K\text{a}$$

What about amine bases? In what form do they exist at physiological pH? As the amine (A^- = RNH_2), or as the ammonium ion (HA = RNH_3^+)? Let's take a 0.0010 M solution of methylamine at pH = 7.3, for example. According to **TABLE 24.1**, the pK_a of methylammonium ion is 10.64, so from the Henderson–Hasselbalch equation, we have

$$\log\frac{[\text{RNH}_2]}{[\text{RNH}_3{}^+]} = \text{pH} - \text{p}K_a = 7.3 - 10.64 = -3.34$$

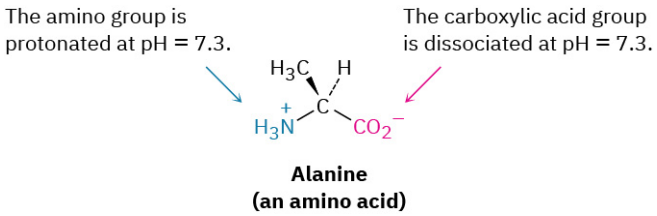
$$\frac{[\text{RNH}_2]}{[\text{RNH}_3{}^+]} = \text{antilog}(-3.34) = 4.6 \times 10^{-4} \quad \text{so} \quad [\text{RNH}_2] = (4.6 \times 10^{-4})[\text{RNH}_3{}^+]$$

In addition, we know that

$$[\text{RNH}_2] + [\text{RNH}_3{}^+] = 0.0010\ \text{M}$$

Solving the two simultaneous equations gives $[RNH_3^+]$ = 0.0010 M and $[RNH_2] = 5 \times 10^{-7}$ M. In other words, at a physiological pH of 7.3, essentially 100% of the methylamine in a 0.0010 M solution exists in its protonated form as methylammonium ion. The same is true of other amine bases, so we always write cellular amines in their protonated form and amino acids in their ammonium carboxylate form to reflect their structures at

physiological pH.

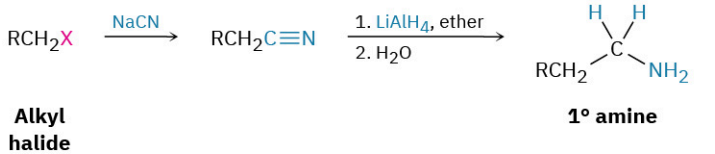


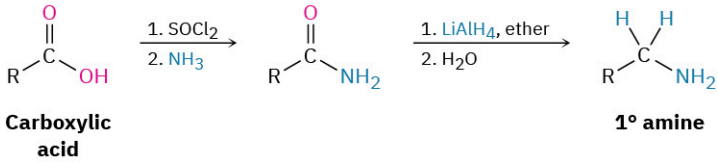
PROBLEM 24-7 Calculate the percentages of neutral and protonated forms present in a solution of 0.0010 Molar pyrimidine at pH = 7.3. The pK_a of pyrimidinium ion is 1.3.

24.6 Synthesis of Amines

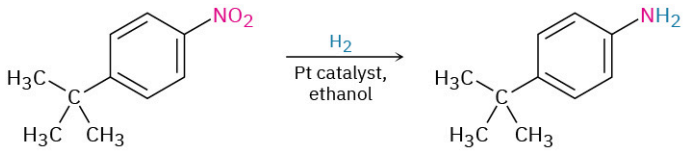
Reduction of Nitriles, Amides, and Nitro Compounds

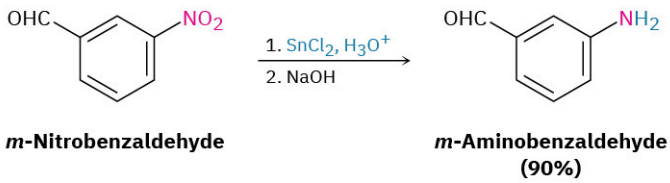
We've already seen in **Section 20.7** and **Section 21.7** how amines can be prepared by reduction of nitriles and amides with $LiAlH_4$. The two-step sequence of S_N2 displacement with CN^- followed by reduction thus converts an alkyl halide into a primary alkylamine having an additional carbon atom. Amide reduction converts carboxylic acids and their derivatives into amines with the same number of carbon atoms.





Arylamines are usually prepared by nitration of an aromatic starting material, followed by reduction of the nitro group (**Section 16.2**). The reduction step can be carried out in many different ways, depending on the circumstances. Catalytic hydrogenation over platinum works well but is often incompatible with the presence elsewhere in the molecule of other reducible groups, such as C=C bonds or carbonyl groups. Iron, zinc, tin, and tin(II) chloride ($SnCl_2$) are also effective when used in acidic aqueous solution. Tin(II) chloride is particularly mild and is often used when other reducible functional groups are present.





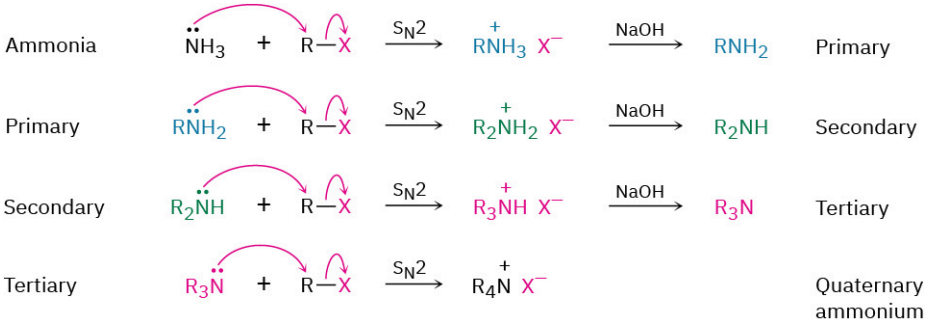
PROBLEM 24-8 Propose structures for either a nitrile or an amide that might be a precursor of each of the following amines:

(a) $CH_3CH_2CH_2NH_2$ **(b)** $(CH_3CH_2CH_2)_2NH$ **(c)** Benzylamine, $C_6H_5CH_2NH_2$

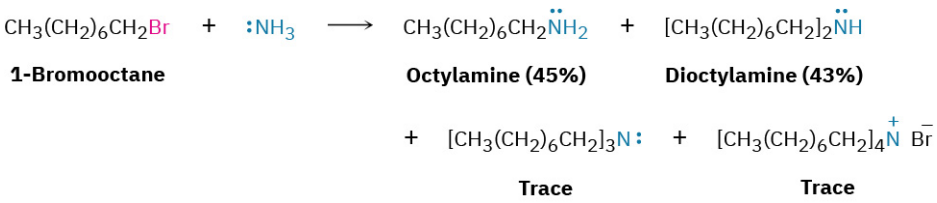
(d) *N*-Ethylaniline

S_N2 Reactions of Alkyl Halides

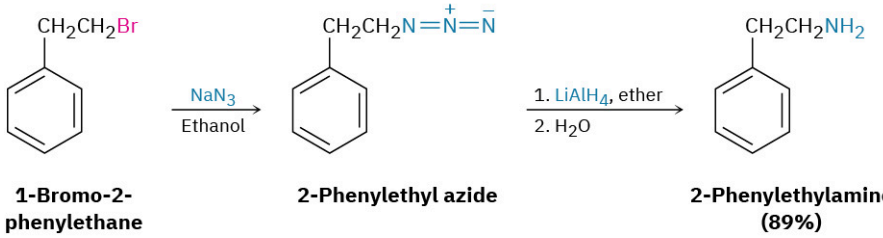
Ammonia and other amines are good nucleophiles in S_N2 reactions. As a result, the simplest method of alkylamine synthesis is by S_N2 alkylation of ammonia or an alkylamine with an alkyl halide. If ammonia is used, a primary amine results; if a primary amine is used, a secondary amine results; and so on. Even tertiary amines react rapidly with alkyl halides to yield quaternary ammonium salts, $R_4N^+ X^-$.



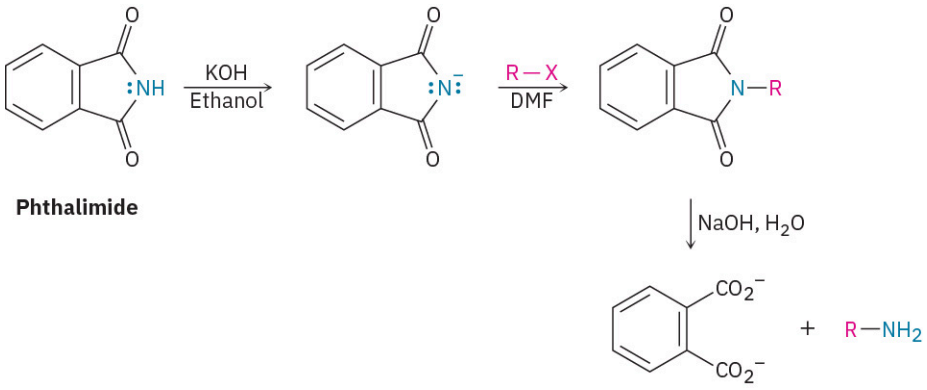
Unfortunately, these reactions don't stop cleanly after a single alkylation has occurred. Because ammonia and primary amines have similar reactivity, the initially formed monoalkylated substance often undergoes further reaction to yield a mixture of products. Even secondary and tertiary amines undergo further alkylation, although to a lesser extent. For example, treatment of 1-bromooctane with a twofold excess of ammonia leads to a mixture containing only 45% octylamine. A nearly equal amount of dioctylamine is produced by double alkylation, along with smaller amounts of trioctylamine and tetraoctylammonium bromide.



A better method for preparing primary amines is to use azide ion, N_3^-, rather than ammonia, as the nucleophile for S_N2 reaction with a primary or secondary alkyl halide. The product is an alkyl azide, which is not nucleophilic, so overalkylation can't occur. Subsequent reduction of the alkyl azide with $LiAlH_4$ then leads to the desired primary amine. Although this method works well, low-molecular-weight alkyl azides are explosive and must be handled carefully.



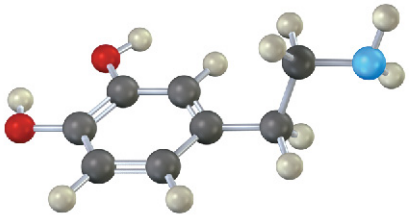
Another alternative for preparing a primary amine from an alkyl halide is the **Gabriel amine synthesis**, which uses a *phthalimide* alkylation. An **imide** (**O=C–N–C=O**) is similar to a *β*-keto ester in that the acidic N–H hydrogen is flanked by two carbonyl groups. Thus, imides are deprotonated by such bases as KOH, and the resultant anions are readily alkylated in a reaction similar to acetoacetic ester synthesis (Section 22.7). Basic hydrolysis of the *N*-alkylated imide then yields a primary amine product. The imide hydrolysis step is analogous to the hydrolysis of an amide (Section 21.7).



Phthalimide

PROBLEM 24-9 Write the mechanism of the last step in Gabriel amine synthesis, the base-promoted hydrolysis of a phthalimide to yield an amine plus phthalate ion.

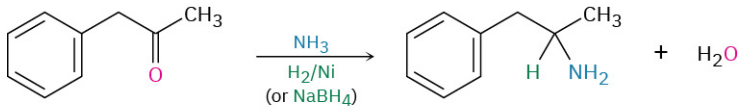
PROBLEM 24-10 Show two methods for the synthesis of dopamine, a neurotransmitter involved in regulation of the central nervous system. Use any alkyl halide needed.



Dopamine

Reductive Amination of Aldehydes and Ketones

Amines can be synthesized in a single step by treatment of an aldehyde or ketone with ammonia or an amine in the presence of a reducing agent, a process called **reductive amination**. For example, amphetamine, a central nervous system stimulant, is prepared commercially by reductive amination of phenyl-2-propanone with ammonia using hydrogen gas over a nickel catalyst as the reducing agent. In the laboratory, either $NaBH_4$ or the related $NaBH(OAc)_3$ is commonly used (OAc = acetate).

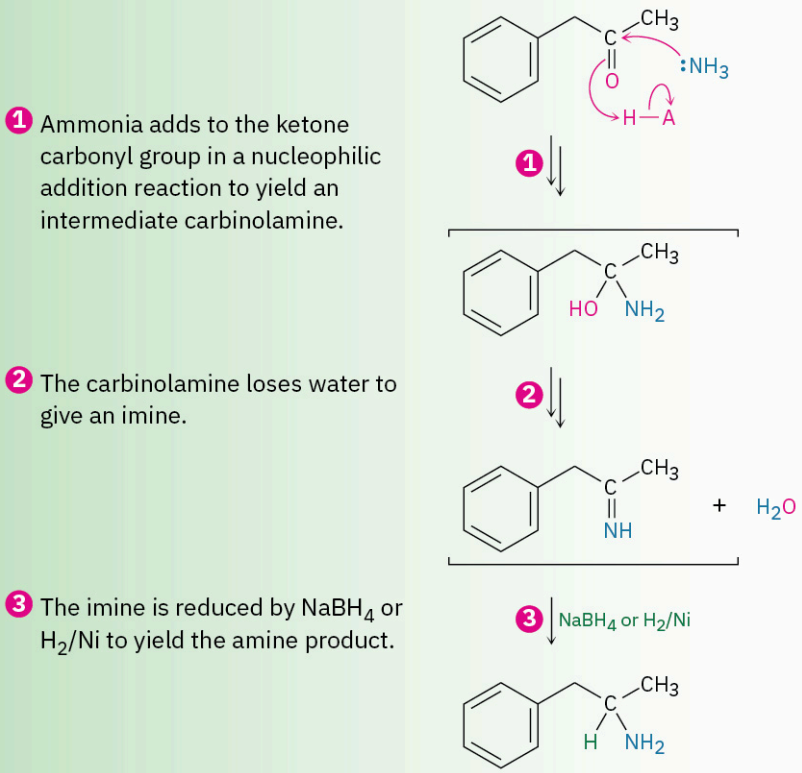


Phenyl-2-propanone **Amphetamine**

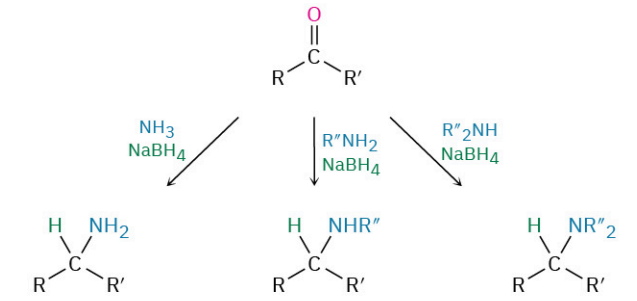
Reductive amination takes place by the pathway shown in FIGURE 24.6. An imine intermediate is first formed by a nucleophilic addition reaction (Section 19.8), and the C═N bond of the imine is then reduced to the amine, much as the C═O bond of a ketone can be reduced to an alcohol.

FIGURE 24.6 MECHANISM

Mechanism for reductive amination of a ketone to yield an amine. Details of the imine-forming step are shown in **FIGURE 19.7**.

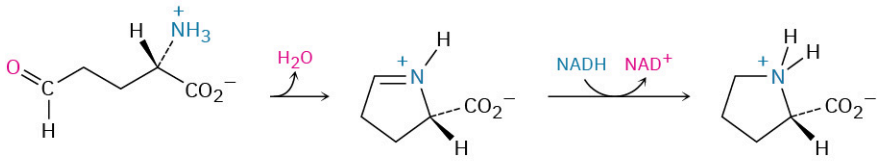


Ammonia, primary amines, and secondary amines can all be used in the reductive amination reaction, yielding primary, secondary, and tertiary amines, respectively.



Primary amine **Secondary amine** **Tertiary amine**

Reductive aminations also occur in various biological pathways. In the biosynthesis of the amino acid proline, for instance, glutamate 5-semialdehyde undergoes internal imine formation to give 1-pyrrolinium 5-carboxylate, which is then reduced by nucleophilic addition of hydride ion to the C=N bond. Reduced nicotinamide adenine dinucleotide, NADH, acts as the biological reducing agent.

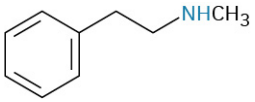


Glutamate 5-semialdehyde **1-Pyrrolinium 5-carboxylate** **Proline**

WORKED EXAMPLE 24.1

Using a Reductive Amination Reaction

How might you prepare *N*-methyl-2-phenylethylamine using a reductive amination reaction?

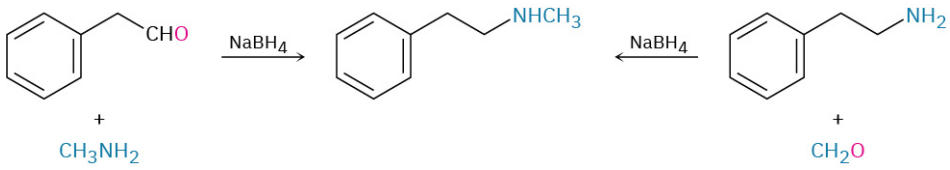


***N*-Methyl-2-phenylethylamine**

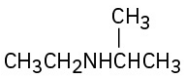
Strategy

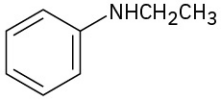
Look at the target molecule, and identify the groups attached to nitrogen. One of the groups must be derived from the aldehyde or ketone component, and the other must be derived from the amine component. In the case of *N*-methyl-2-phenylethylamine, two combinations can lead to the product: phenylacetaldehyde plus methylamine or formaldehyde plus 2-phenylethylamine. It's usually better to choose the combination with the simpler amine component—methylamine in this case—and to use an excess of that amine as reactant.

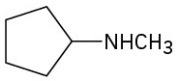
Solution



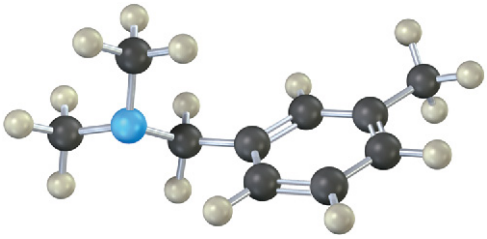
PROBLEM 24-11 How might the following amines be prepared using reductive amination reactions? Show all precursors if more than one is possible.

(a) 

(b) 

(c) 

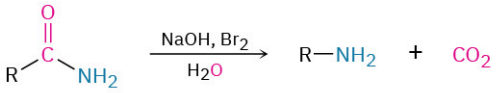
PROBLEM 24-12 How could you prepare the following amine using a reductive amination reaction?



Hofmann and Curtius Rearrangements

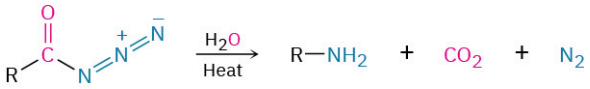
Carboxylic acid derivatives can be converted into primary amines with loss of one carbon atom by both the **Hofmann rearrangement** and the **Curtius rearrangement**. Although the Hofmann rearrangement involves a primary amide and the Curtius rearrangement involves an acyl azide, both proceed through similar mechanisms.

Hofmann rearrangement



An amide

Curtius rearrangement

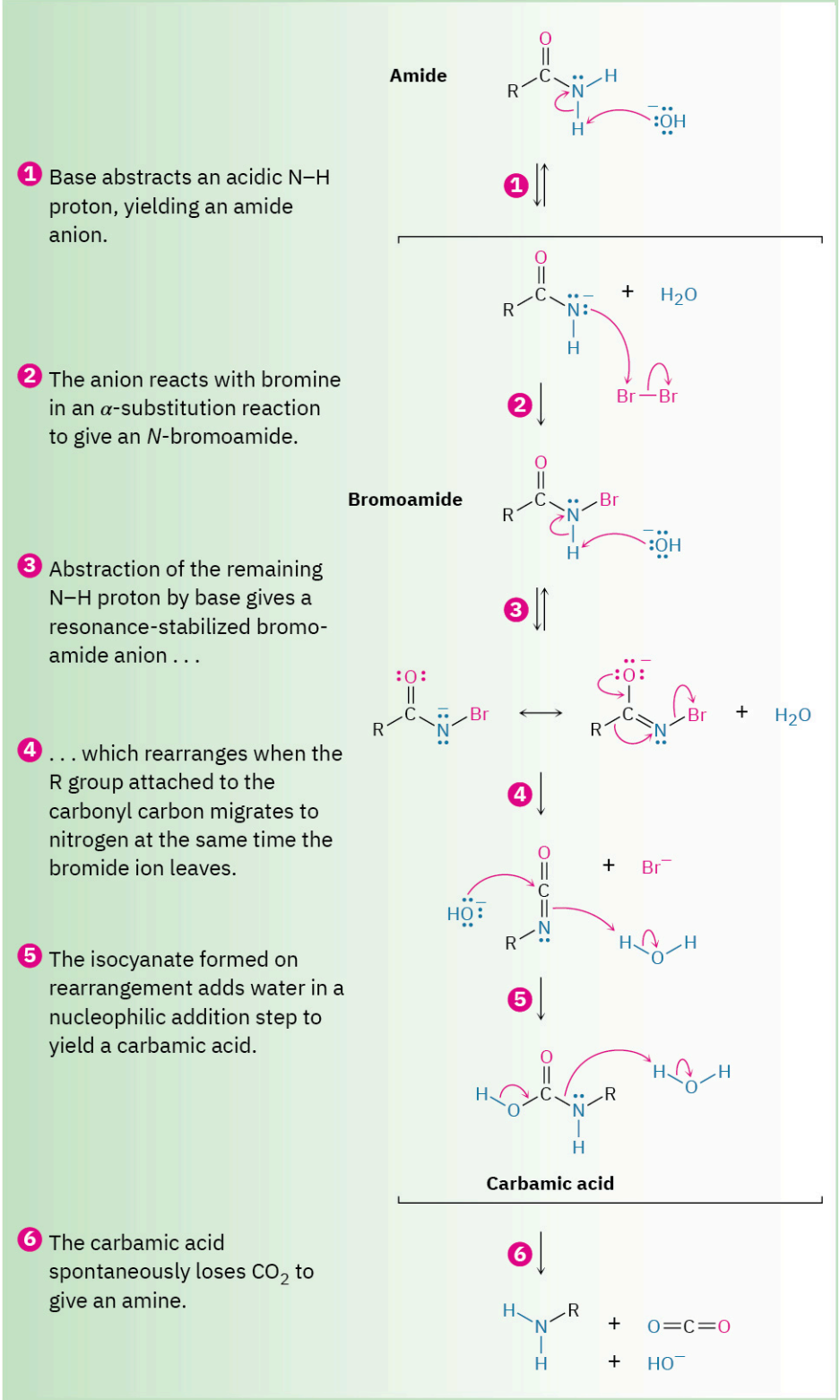


An acyl azide

Hofmann rearrangement occurs when a primary amide, $RCONH_2$, is treated with Br_2 and base (**FIGURE 24.7**). The overall mechanism is lengthy, but most of the individual steps have been encountered before. Thus, the bromination of an amide in steps **1** and **2** is analogous to the base-promoted bromination of a ketone enolate ion (**Section 22.6**), and the rearrangement of the bromoamide anion in step **4** is analogous to a carbocation rearrangement (**Section 7.11**). Nucleophilic addition of water to the isocyanate carbonyl group in step **5** is a typical carbonyl-group process (**Section 19.4**), as is the final decarboxylation step **6** (**Section 22.7**).

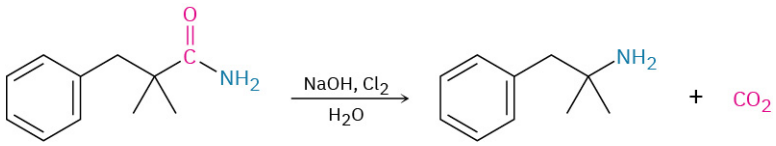
FIGURE 24.7 MECHANISM

Mechanism of the Hofmann rearrangement of an amide to an amine. Each step is analogous to a reaction studied previously.



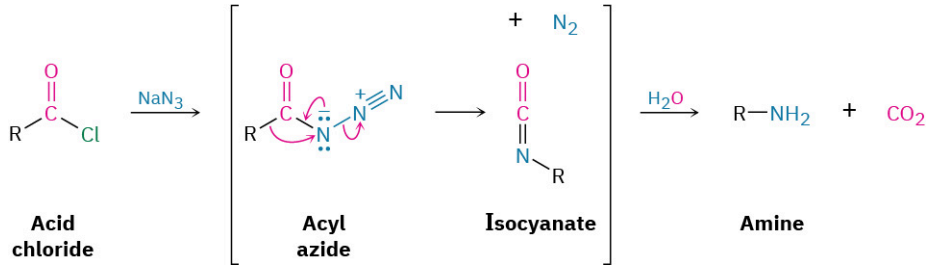
Despite its mechanistic complexity, the Hofmann rearrangement often gives high yields of both arylamines and

alkylamines. For example, the appetite-suppressant drug phentermine is prepared commercially by Hofmann rearrangement of a primary amide. Commonly known by the name Fen-Phen, the combination of phentermine with another appetite-suppressant, fenfluramine, is suspected of causing heart damage.



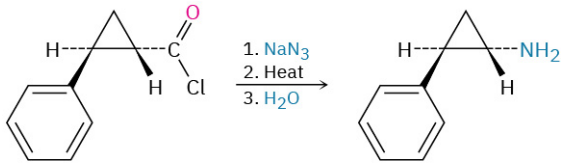
2,2-Dimethyl-3-phenylpropanamide **Phentermine**

The Curtius rearrangement, like the Hofmann rearrangement, involves migration of an –R group from the C=O carbon atom to the neighboring nitrogen with simultaneous loss of a leaving group. The reaction takes place on heating an acyl azide that is itself prepared by nucleophilic acyl substitution of an acid chloride.



Acid chloride **Acyl azide** **Isocyanate** **Amine**

Also like the Hofmann rearrangement, the Curtius rearrangement is often used commercially. The antidepressant drug tranylcypromine, for instance, is made by Curtius rearrangement of 2-phenylcyclopropanecarbonyl chloride.



***trans*-2-Phenylcyclopropanecarbonyl chloride** **Tranylcypromine**

WORKED EXAMPLE 24.2

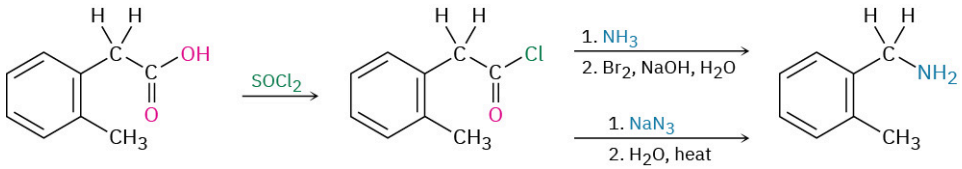
Using the Hofmann and Curtius Rearrangements

How would you prepare *o*-methylbenzylamine from a carboxylic acid, using both Hofmann and Curtius rearrangements?

Strategy

Both Hofmann and Curtius rearrangements convert a carboxylic acid derivative—either an amide (Hofmann) or an acid chloride (Curtius)—into a primary amine with loss of one carbon, RCOY → RNH_2. Both reactions begin with the same carboxylic acid, which can be identified by replacing the $–NH_2$ group of the amine product by a $–CO_2H$ group. In the present instance, *o*-methylphenylacetic acid is needed.

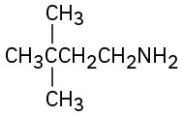
Solution



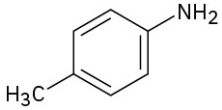
***o*-Methylphenylacetic acid** ***o*-Methylbenzylamine**

PROBLEM 24-13 How would you prepare the following amines, using both Hofmann and Curtius rearrangements on a carboxylic acid derivative?

(a)



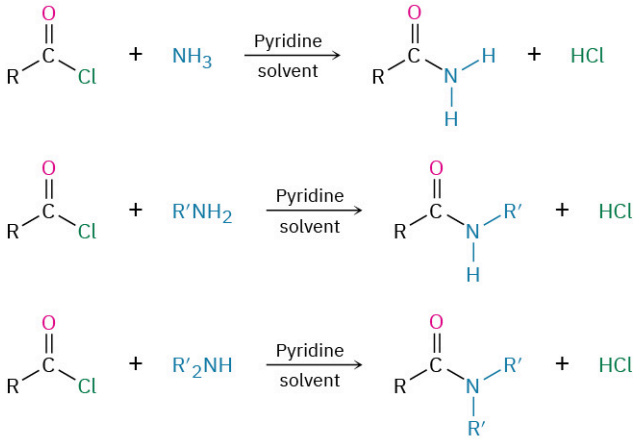
(b)



24.7 Reactions of Amines

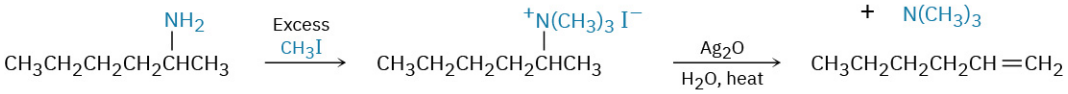
Alkylation and Acylation

We've already studied the two most general reactions of amines—alkylation and acylation. As we saw earlier in this chapter, primary, secondary, and tertiary amines can be alkylated by reaction with a primary alkyl halide. Alkylations of primary and secondary amines are difficult to control and often give mixtures of products, but tertiary amines are cleanly alkylated to give quaternary ammonium salts. Primary and secondary (but not tertiary) amines can also be acylated by nucleophilic acyl substitution reaction with an acid chloride or an acid anhydride to yield an amide (**Section 21.4** and **Section 21.5**). Note that overacylation of the nitrogen does not occur because the amide product is much less nucleophilic and less reactive than the starting amine.



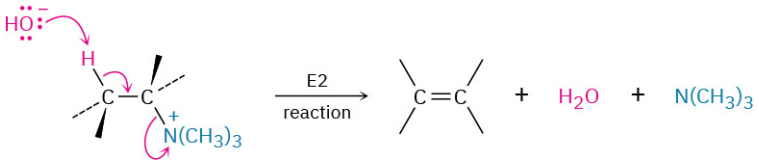
Hofmann Elimination Reaction

Like alcohols, amines can be converted into alkenes by an elimination reaction. But because an amide ion, NH_2^-, is such a poor leaving group, it must first be converted into a better leaving group. In the **Hofmann elimination reaction**, an amine is completely methylated by reaction with an excess amount of iodomethane to produce the corresponding quaternary ammonium salt. This salt then undergoes elimination to give an alkene on heating with a base, typically silver oxide, Ag_2O. For example, 1-methylpentylamine is converted into 1-hexene.



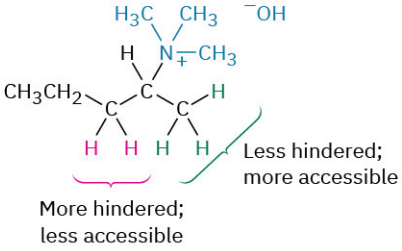
1-Methylpentylamine | **(1-Methylpentyl)trimethylammonium iodide** | **1-Hexene (60%)**

Silver oxide acts by exchanging iodide ion for hydroxide ion in the quaternary salt, thus providing the base necessary for elimination. The actual elimination step is an E2 reaction (**Section 11.8**) in which hydroxide ion removes a proton while the positively charged nitrogen atom leaves.

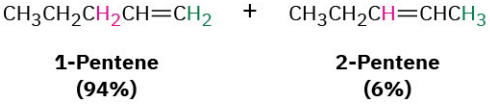


Quaternary ammonium salt **Alkene**

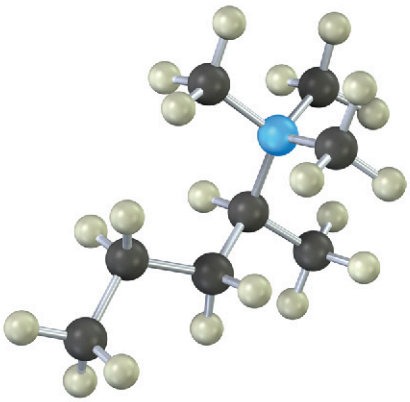
Unlike what happens in other E2 reactions, the major product of Hofmann elimination is the less highly substituted alkene rather than the more highly substituted one, as shown by the reaction of (1-methylbutyl)trimethylammonium hydroxide to give 1-pentene rather than the alternative 2-pentene. The reason for this non-Zaitsev result is probably steric. Because of the large size of the trialkylamine leaving group, the base must abstract a hydrogen from the more accessible, least hindered position.



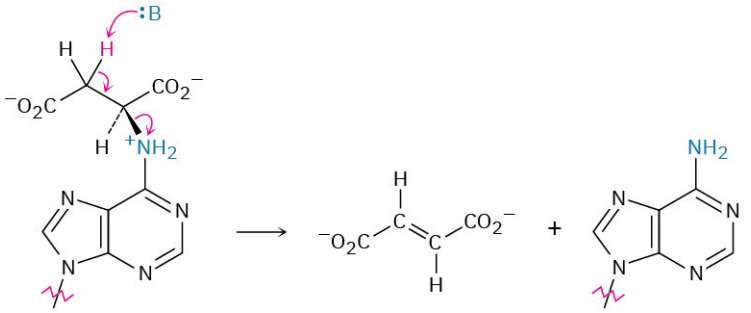
(1-Methylbutyl)trimethylammonium hydroxide



1-Pentene (94%) **2-Pentene (6%)**



The Hofmann elimination reaction is not often currently used in the laboratory, but analogous biological eliminations occur frequently, although usually with protonated ammonium ions rather than quaternary ammonium salts. In the biosynthesis of nucleic acids, for instance, a substance called adenylosuccinate undergoes an elimination of a positively charged nitrogen to give fumarate plus adenosine monophosphate.

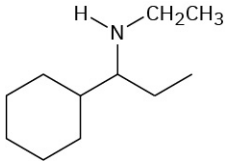


Adenylosuccinate **Fumarate** **Adenosine monophosphate**

WORKED EXAMPLE 24.3

Predicting the Product of a Hofmann Elimination

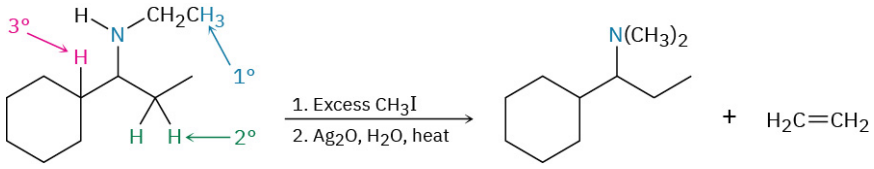
What product would you expect from Hofmann elimination of the following amine?



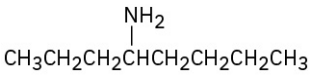
Strategy

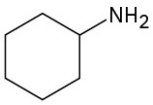
The Hofmann elimination is an E2 reaction that converts an amine into an alkene and occurs with non-Zaitsev regiochemistry to form the less highly substituted double bond. To predict the product, look at the reactant and identify the positions from which elimination might occur (the positions two carbons away from nitrogen). Then, carry out an elimination using the most accessible hydrogen. In the present instance, there are three possible positions from which elimination might occur—one primary, one secondary, and one tertiary. The primary position is the most accessible and leads to the least highly substituted alkene, ethylene.

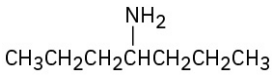
Solution

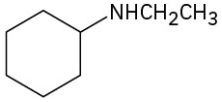


PROBLEM 24-14 What products would you expect from Hofmann elimination of the following amines? If more than one product is formed, indicate which is major.

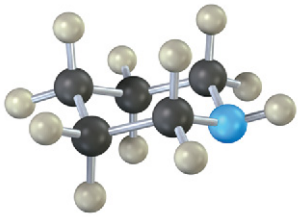
(a) 

(b) 

(c) 

(d) 

PROBLEM 24-15 What product would you expect from Hofmann elimination of a heterocyclic amine such as piperidine? Write all the steps.

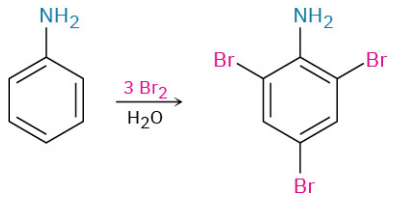


Piperidine

24.8 Reactions of Arylamines

Electrophilic Aromatic Substitutions

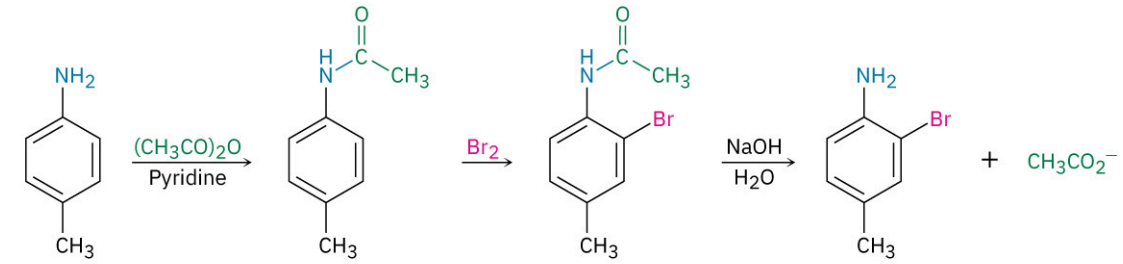
An amino group is strongly activating and ortho- and para-directing in electrophilic aromatic substitution reactions (**Section 16.4**). This high reactivity of amino-substituted benzenes can be a drawback at times because it's often difficult to prevent polysubstitution. Reaction of aniline with Br_2, for instance, takes place rapidly and yields the 2,4,6-tribrominated product. The amino group is so strongly activating that it's not possible to stop at the monobromo stage.



Aniline **2,4,6-Tribromoaniline (100%)**

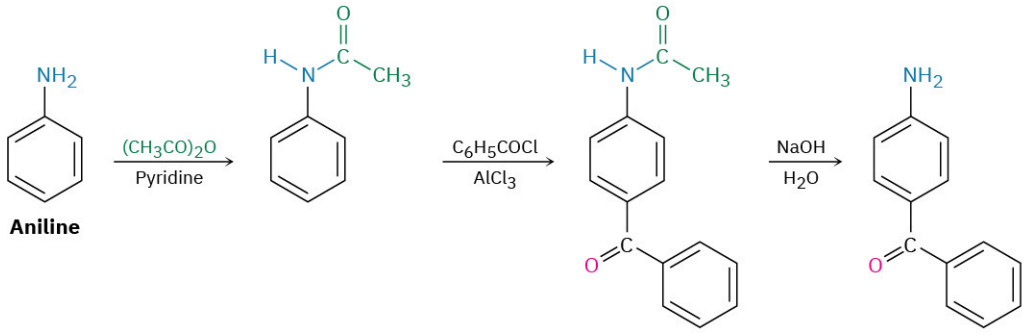
Another drawback to the use of amino-substituted benzenes in electrophilic aromatic substitution reactions is that Friedel–Crafts reactions are not successful (**Section 16.3**). The amino group forms an acid–base complex with the $AlCl_3$ catalyst, which prevents further reaction. Both drawbacks can be overcome, however, by carrying out electrophilic aromatic substitution reactions on the corresponding amide rather than on the free amine.

As we saw in **Section 21.5**, treatment of an amine with acetic anhydride yields the corresponding acetyl amide, or acetamide. Although still activating and ortho-, para-directing, amido substituents (R–N–C=O) are less strongly activating and less basic than amino groups because their nitrogen lone-pair electrons are delocalized by the neighboring carbonyl group. As a result, bromination of an *N*-arylamide occurs cleanly to give a monobromo product, and hydrolysis of the amide with aqueous base then gives the free amine. For example, *p*-toluidine (4-methylaniline) can be acetylated, brominated, and hydrolyzed to yield 2-bromo-4-methylaniline. None of the 2,6-dibrominated product is obtained.



***p*-Toluidine** **2-Bromo-4-methyl-aniline (79%)**

Friedel–Crafts alkylations and acylations of *N*-arylamides also proceed normally. For example, benzoylation of acetanilide (*N*-acetylaniline) under Friedel–Crafts conditions gives 4-aminobenzophenone in 80% yield after hydrolysis.

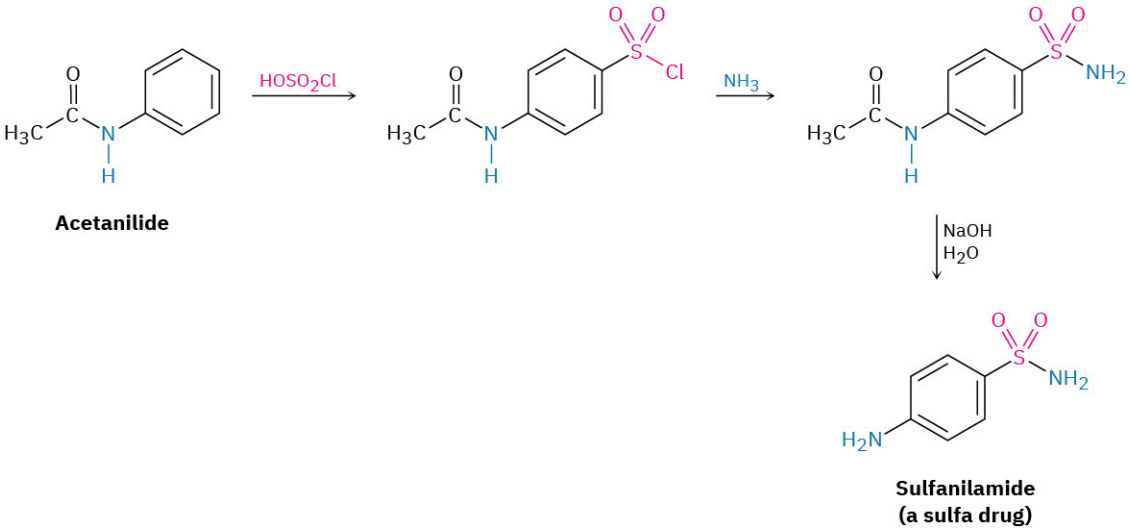


Aniline **4-Aminobenzophenone (80%)**

Moderating the reactivity of an amino-substituted benzene by forming an amide is a useful trick that allows many kinds of electrophilic aromatic substitutions to be carried out that would otherwise be impossible. One example is the preparation of the sulfa drugs, such as sulfanilamide.

Sulfa drugs were among the first pharmaceutical agents to be used clinically against bacterial infection. Although they have largely been replaced today by safer and more powerful antibiotics, sulfa drugs are credited with saving the lives of thousands of wounded during World War II and are still prescribed for urinary tract infections. They are prepared by chlorosulfonation of acetanilide, followed by reaction of

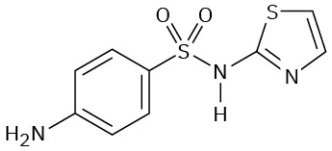
p-(*N*-acetylamino)benzenesulfonyl chloride with ammonia or some other amine to give a sulfonamide. Hydrolysis of the amide then yields the sulfa drug. Note that hydrolysis of the amide can be carried out in the presence of the sulfonamide group because sulfonamides hydrolyze very slowly.



Acetanilide

Sulfanilamide (a sulfa drug)

PROBLEM 24-16 Propose a synthesis of the drug sulfathiazole from benzene and any necessary amine.

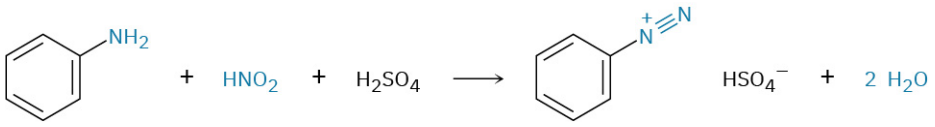


Sulfathiazole

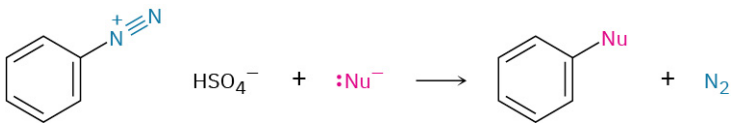
PROBLEM 24-17 Propose syntheses of the following compounds from benzene: **(a)** *N,N*-Dimethylaniline **(b)** *p*-Chloroaniline **(c)** *m*-Chloroaniline **(d)** 2,4-Dimethylaniline

Diazonium Salts: The Sandmeyer Reaction

Primary arylamines react with nitrous acid, HNO_2, to yield stable *arenediazonium salts*, $Ar{-}\overset{+}{N}{\equiv}N\ X^-$, a process called a *diazotization* reaction. Alkylamines also react with nitrous acid, but the corresponding alkanediazonium products are so reactive they can't be isolated. Instead, they lose nitrogen instantly to yield carbocations. The analogous loss of N_2 from an arenediazonium ion to yield an aryl cation is disfavored by the instability of the cation.



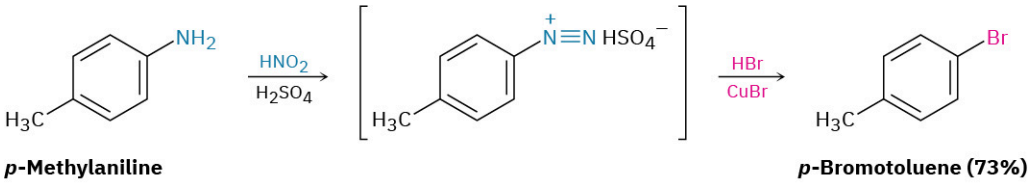
Arenediazonium salts are useful because the diazonio group (–N≡N) can be replaced by a nucleophile in a substitution reaction.



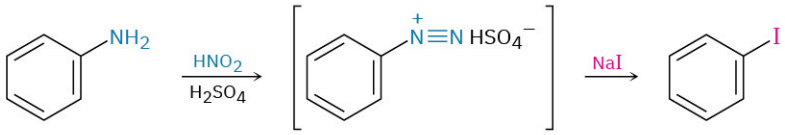
Many different nucleophiles—halide, hydride, cyanide, and hydroxide among others—react with arenediazonium salts, yielding many different kinds of substituted benzenes. The overall sequence of (**1**) nitration, (**2**) reduction, (**3**) diazotization, and (**4**) nucleophilic substitution is perhaps the single most versatile method of aromatic substitution.

Aryl chlorides and bromides are prepared by reaction of an arenediazonium salt with the corresponding copper(I) halide, CuX, a process called the **Sandmeyer reaction**. Aryl iodides can be prepared by direct reaction

with NaI without using a copper(I) salt. Yields generally fall between 60% and 80%.

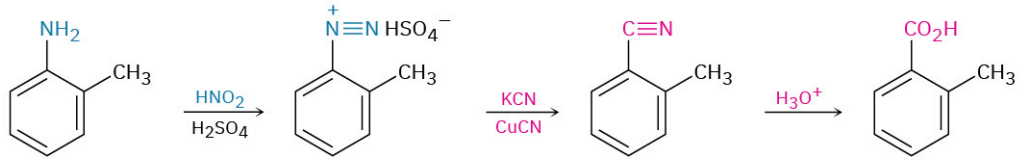


***p*-Methylaniline** **p-Bromotoluene (73%)**



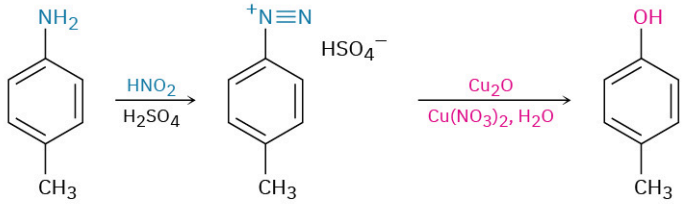
Aniline **Iodobenzene (67%)**

Similar treatment of an arenediazonium salt with CuCN yields the nitrile ArCN, which can then be further converted into other functional groups such as carboxyl. For example, Sandmeyer reaction of *o*-methylbenzenediazonium bisulfate with CuCN yields *o*-methylbenzonitrile, which can be hydrolyzed to give *o*-methylbenzoic acid. This product can't be prepared from *o*-xylene by the usual side-chain oxidation route because both methyl groups would be oxidized.



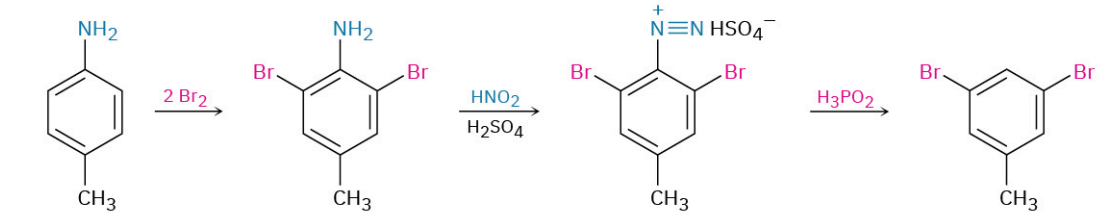
***o*-Methylaniline** ***o*-Methylbenzene-diazonium bisulfate** ***o*-Methylbenzonitrile** ***o*-Methylbenzoic acid**

The diazonio group can also be replaced by –OH to yield a phenol and by –H to yield an arene. A phenol is prepared by reaction of the arenediazonium salt with copper(I) oxide in an aqueous solution of copper(II) nitrate, a reaction that is especially useful because few other general methods exist for introducing an –OH group onto an aromatic ring.

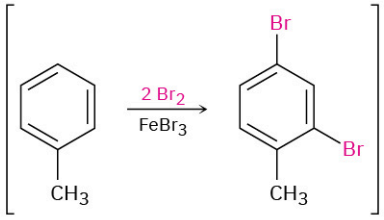


***p*-Methylaniline (*p*-Toluidine)** ***p*-Cresol (93%)**

Reduction of a diazonium salt to give an arene occurs on treatment with hypophosphorous acid, H_3PO_2. This reaction is used primarily when there is a need for temporarily introducing an amino substituent onto a ring to take advantage of its directing effect. Suppose, for instance, that you needed to make 3,5-dibromotoluene. This product can't be made by direct bromination of toluene because reaction would occur at positions 2 and 4. Starting with *p*-methylaniline (*p*-toluidine), however, dibromination occurs ortho to the strongly directing amino substituent, and diazotization followed by treatment with H_3PO_2 to remove the amino group yields the desired product.

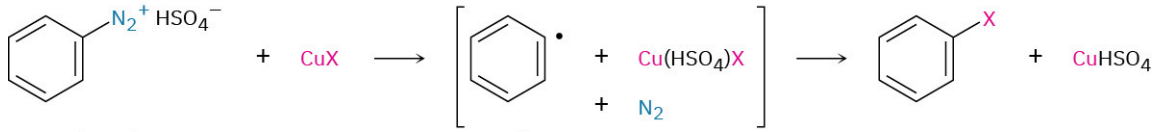


***p*-Methylaniline** **3,5-Dibromotoluene**



Toluene **2,4-Dibromotoluene**

Mechanistically, these diazonio replacement reactions occur through radical rather than polar pathways. In the presence of a copper(I) compound, for instance, it's thought that the arenediazonium ion is first converted to an aryl radical plus copper(II), followed by subsequent reaction to give product plus regenerated copper(I) catalyst.

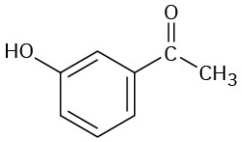


Diazonium compound **Aryl radical**

WORKED EXAMPLE 24.4

Using Diazonium Replacement Reactions

How would you prepare *m*-hydroxyacetophenone from benzene, using a diazonium replacement reaction in your scheme?



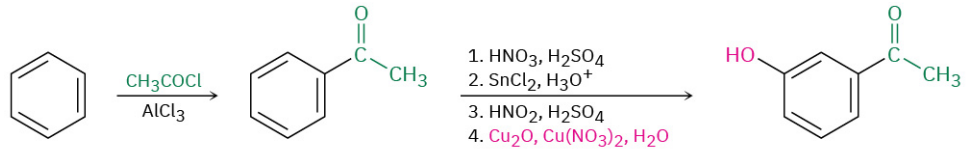
***m*-Hydroxyacetophenone**

Strategy

As always, organic syntheses are planned by working retrosynthetically from the final product, one step at a time. First, identify the functional groups in the product and recall how those groups can be synthesized. *m*-Hydroxyacetophenone has an –OH group and a $-COCH_3$ group in a meta relationship on a benzene ring. A hydroxyl group is generally introduced onto an aromatic ring by a four-step sequence of nitration, reduction, diazotization, and diazonio replacement. An acetyl group is introduced by a Friedel–Crafts acylation reaction.

Next, ask yourself what an immediate precursor of the target might be. Since an acetyl group is a meta director while a hydroxyl group is an ortho and para director, acetophenone might be a precursor of *m*-hydroxyacetophenone. Benzene, in turn, is a precursor of acetophenone.

Solution

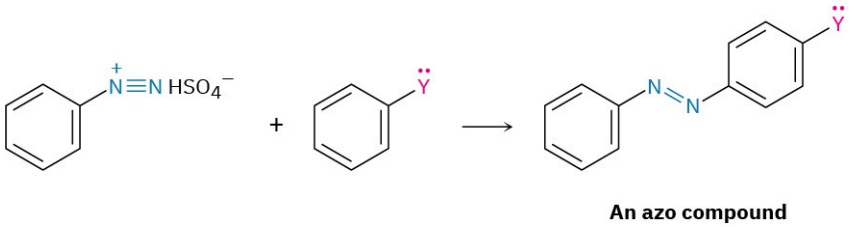


PROBLEM 24-18 How would you prepare the following compounds from benzene, using a diazonium replacement reaction in your scheme?

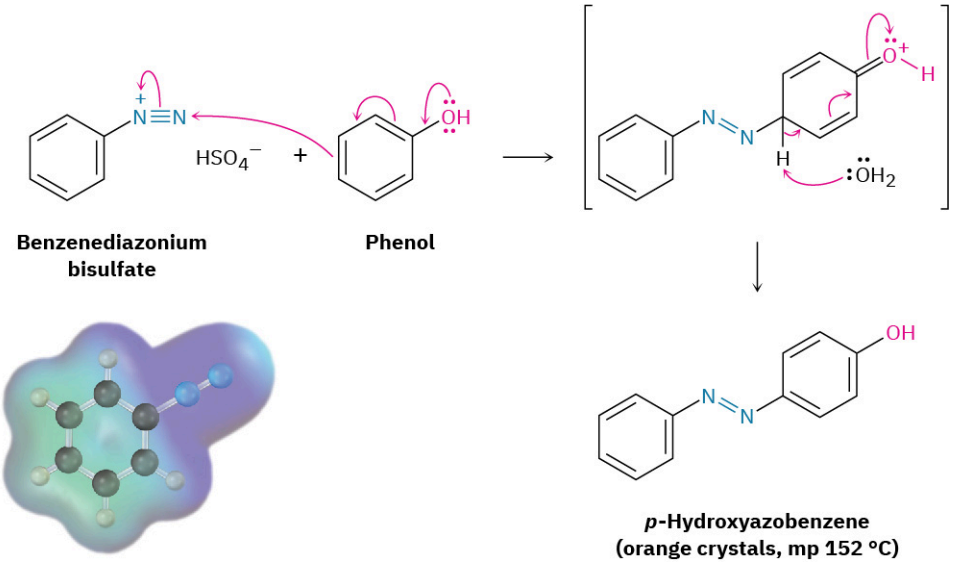
(a) *p*-Bromobenzoic acid **(b)** *m*-Bromobenzoic acid **(c)** *m*-Bromochlorobenzene
(d) *p*-Methylbenzoic acid **(e)** 1,2,4-Tribromobenzene

Diazonium Coupling Reactions

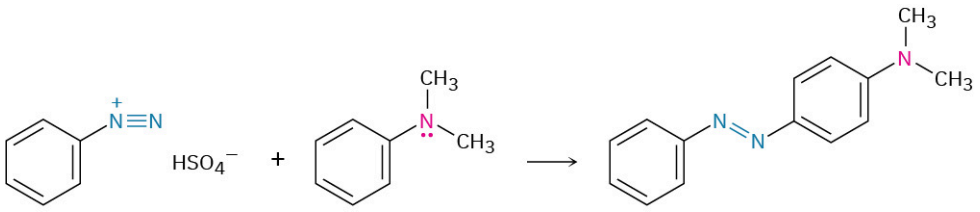
Arenediazonium salts undergo a coupling reaction with activated aromatic rings such as phenols and arylamines to yield brightly colored **azo compounds, Ar–N=N–Ar′**.



Diazonium coupling reactions are typical electrophilic aromatic substitutions in which the positively charged diazonium ion is the electrophile that reacts with the electron-rich ring of a phenol or arylamine. Reaction usually occurs at the para position.



Azo-coupled products are widely used as dyes for textiles because their extended conjugated π electron system causes them to absorb in the visible region of the electromagnetic spectrum (Section 14.9). *p*-(Dimethylamino)-azobenzene, for instance, is a bright yellow compound that was at one time used as a coloring agent in margarine.

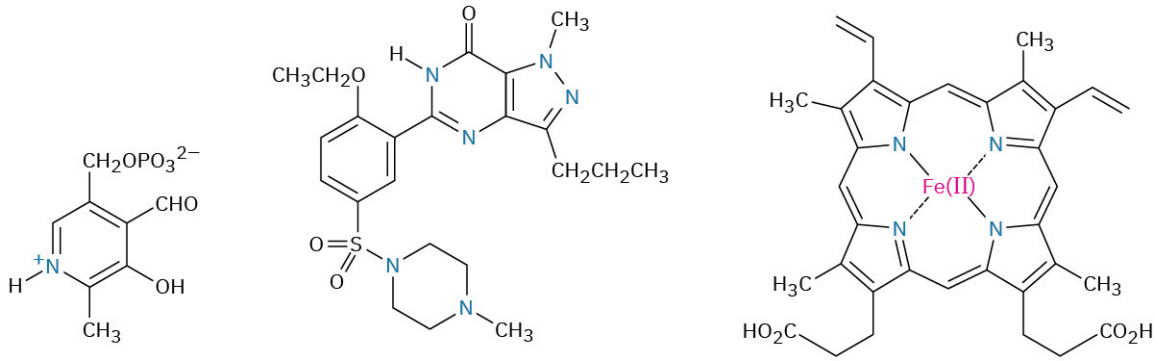


Benzenediazonium bisulfate **N,N-Dimethylaniline** ***p*-(Dimethylamino)azobenzene (yellow crystals, mp 127 °C)**

PROBLEM 24-19 Propose a synthesis of *p*-(dimethylamino)azobenzene with benzene as your only organic starting material.

24.9 Heterocyclic Amines

As noted in **Section 15.5** in connection with a discussion of aromaticity, a cyclic organic compound that contains atoms of two or more elements in its ring is a called a *heterocycle*. Heterocyclic amines are particularly common, and many have important biological properties. Pyridoxal phosphate, a coenzyme; sildenafil (Viagra), a well-known pharmaceutical; and heme, the oxygen carrier in blood, are a few examples.

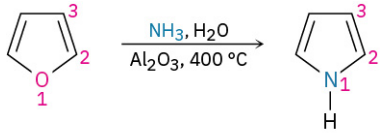


Pyridoxal phosphate (a coenzyme) **Sildenafil (Viagra)** **Heme**

Most heterocycles have the same chemistry as their open-chain counterparts. Lactones and acyclic esters behave similarly, lactams and acyclic amides behave similarly, and cyclic and acyclic ethers behave similarly. In certain cases, however, particularly when the ring is unsaturated, heterocycles have unique and interesting properties.

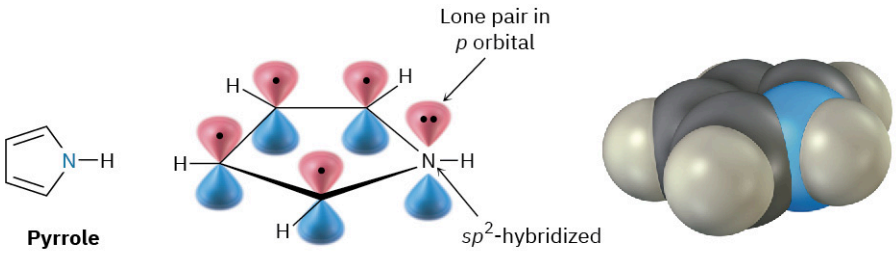
Pyrrole and Imidazole

Pyrrole, the simplest five-membered unsaturated heterocyclic amine, is obtained commercially by treatment of furan with ammonia over an alumina catalyst at 400 °C. Furan, the oxygen-containing analog of pyrrole, is obtained by acid-catalyzed dehydration of the five-carbon sugars found in oat hulls and corncobs.



Furan **Pyrrole**

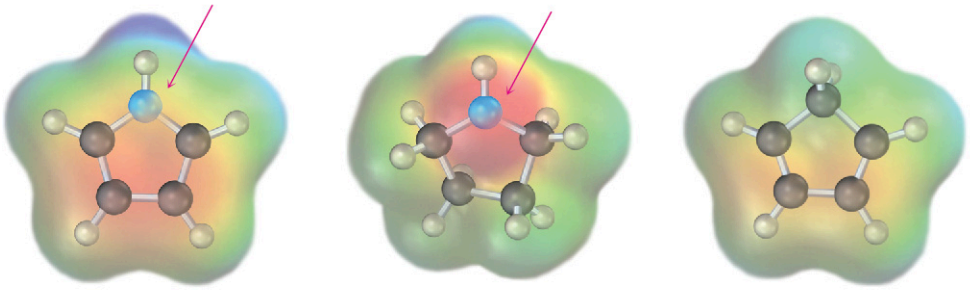
Although pyrrole appears to be both an amine and a conjugated diene, its chemical properties are not consistent with either of these structural features. Unlike most other amines, pyrrole is not basic—the pK_a of the pyrrolinium ion is 0.4; unlike most other conjugated dienes, pyrrole undergoes electrophilic substitution reactions rather than additions. The reason for both of these properties, as noted in **Section 15.5**, is that pyrrole has six π electrons and is aromatic. Each of the four carbons contributes one π electron, and the sp^2-hybridized nitrogen contributes two more from its lone pair.



Pyrrole

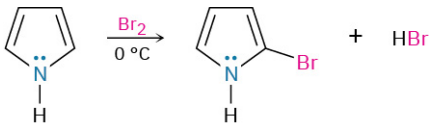
Six π electrons

Because the nitrogen lone pair is a part of the aromatic sextet, protonation on nitrogen would destroy the aromaticity of the ring. The nitrogen atom in pyrrole is therefore less electron-rich, less basic, and less nucleophilic than the nitrogen in an aliphatic amine. By the same token, the carbon atoms of pyrrole are more electron-rich and more nucleophilic than typical double-bond carbons. The pyrrole ring is therefore reactive toward electrophiles in the same way as enamines (Section 23.11). Electrostatic potential maps show how the pyrrole nitrogen is electron-poor (less red) compared with the nitrogen in its saturated counterpart pyrrolidine, while the pyrrole carbon atoms are electron-rich (more red) compared with the carbons in 1,3-cyclopentadiene.



Pyrrole **Pyrrolidine** **1,3-Cyclopentadiene**

The chemistry of pyrrole is similar to that of activated benzene rings. In general, however, the heterocycles are more reactive toward electrophiles than benzene rings, and low temperatures are often necessary to control the reactions. Halogenation, nitration, sulfonation, and Friedel–Crafts acylation can all be accomplished. For example:



Pyrrole **2-Bromopyrrole (92%)**

Electrophilic substitutions normally occur at C2, the position next to the nitrogen, because reaction at this position leads to a more stable intermediate cation having three resonance forms, whereas reaction at C3 gives a less stable cation with only two resonance forms (FIGURE 24.8).

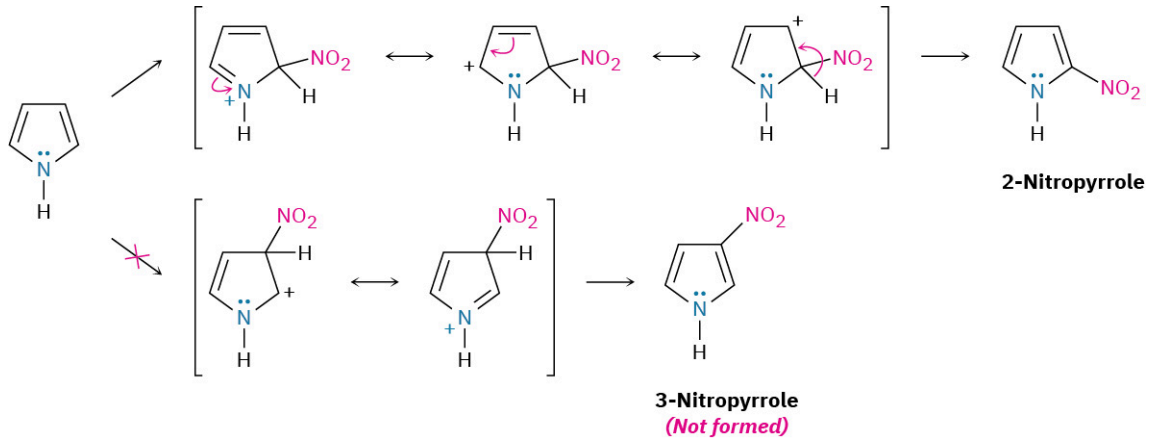
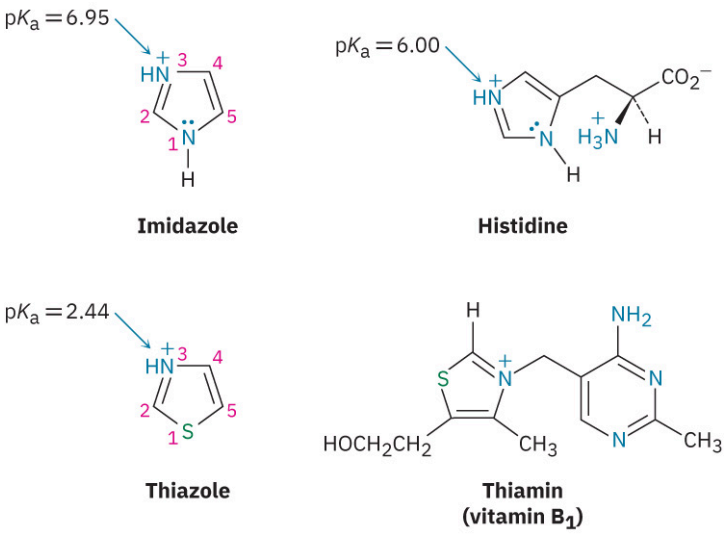


FIGURE 24.8 **Electrophilic nitration of pyrrole.** The intermediate produced by reaction at C2 is more stable than that produced by reaction at C3.

Other common five-membered heterocyclic amines include imidazole and thiazole. Imidazole, a constituent of the amino acid histidine, has two nitrogens, only one of which is basic. Thiazole, the five-membered ring system on which the structure of thiamin (vitamin B_1) is based, also contains a basic nitrogen that is alkylated in thiamin to form a quaternary ammonium ion.



PROBLEM 24-20 Draw an orbital picture of thiazole. Assume that both the nitrogen and sulfur atoms are sp^2-hybridized, and show the orbitals that the lone pairs occupy.

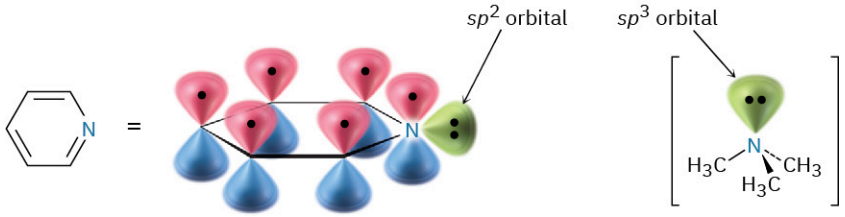
PROBLEM 24-21 What is the percent protonation of the imidazole nitrogen atom in histidine at a physiological pH of 7.3 **(Section 24.5)**?

Pyridine and Pyrimidine

Pyridine is the nitrogen-containing heterocyclic analog of benzene. Like benzene, pyridine is a flat, aromatic molecule, with bond angles of 120° and C–C bond lengths of 139 pm, intermediate between typical single and double bonds. The five carbon atoms and the sp^2-hybridized nitrogen atom each contribute one π electron to the aromatic sextet, and the lone-pair electrons occupy an sp^2 orbital in the plane of the ring (Section 15.5).

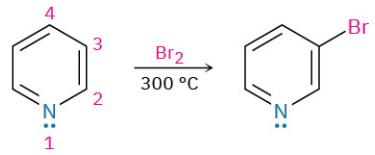
As shown previously in TABLE 24.1, pyridine (pK_a = 5.25) is a stronger base than pyrrole but a weaker base than the alkylamines. The diminished basicity of pyridine compared with that of alkylamines is due to the fact that the lone-pair electrons on the pyridine nitrogen are in an sp^2 orbital, while those on an alkylamine nitrogen are in an sp^3 orbital. Because *s* orbitals have their maximum electron density at the nucleus but *p* orbitals have a node at the nucleus, electrons in an orbital with more *s* character are held more closely to the positively charged nucleus and are less available for bonding. As a result, the sp^2-hybridized nitrogen atom (33% *s* character) in

pyridine is less basic than the sp^3-hybridized nitrogen in an alkylamine (25% *s* character).



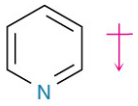
Pyridine

Unlike benzene, pyridine undergoes electrophilic aromatic substitution reactions with difficulty. Halogenation can be carried out under drastic conditions, but nitration occurs in very low yield, and Friedel–Crafts reactions are not successful. Reactions usually give the 3-substituted product.



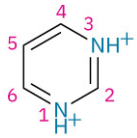
Pyridine **3-Bromopyridine (30%)**

The low reactivity of pyridine toward electrophilic aromatic substitution is caused by a combination of factors. One is that acid–base complexation between the basic ring's nitrogen atom and the incoming electrophile places a positive charge on the ring, thereby deactivating it. Equally important is that the electron density of the ring is decreased by the electron-withdrawing inductive effect of the electronegative nitrogen atom. Thus, pyridine has a substantial dipole moment (μ = 2.26 D), with the ring carbons acting as the positive end of the dipole. Reaction of an electrophile with the positively polarized carbon atoms is therefore difficult.



μ = 2.26 D

In addition to pyridine, the six-membered diamine pyrimidine is also found commonly in biological molecules, particularly as a constituent of nucleic acids. With a pK_a of 1.3, pyrimidine is substantially less basic than pyridine because of the inductive effect of the second nitrogen.

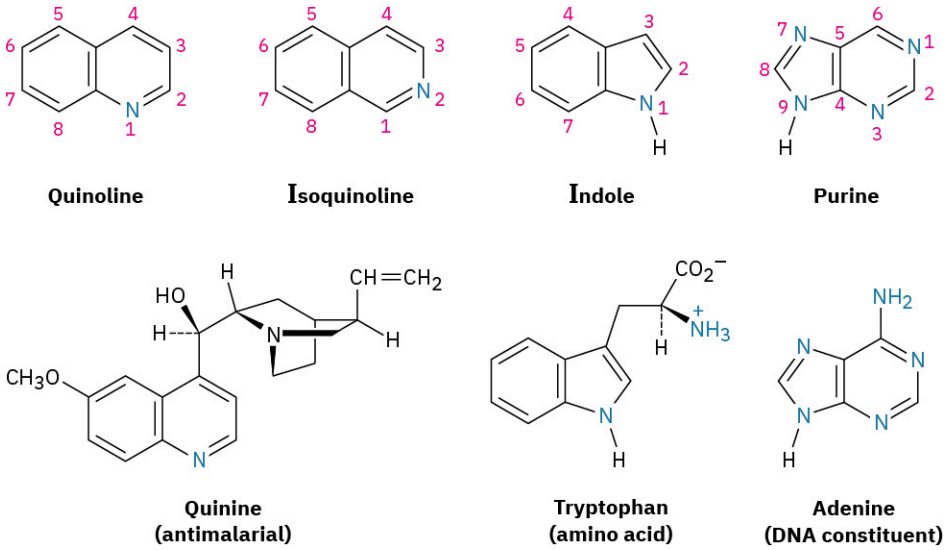


Pyrimidine
pK_a = 1.3

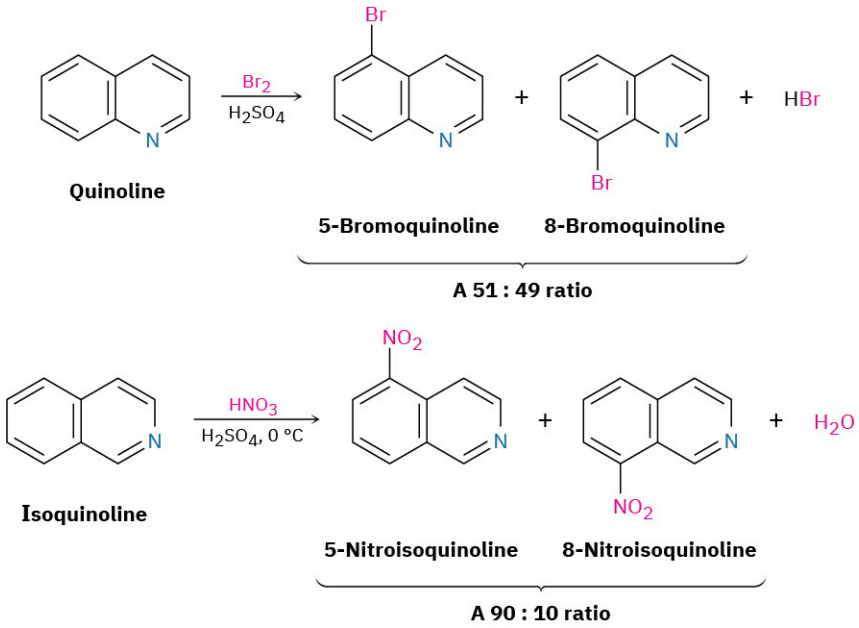
PROBLEM 24-22 Electrophilic aromatic substitution reactions of pyridine normally occur at C3. Draw the carbocation intermediates resulting from reaction of an electrophile at C2, C3, and C4, and explain the observed result.

Polycyclic Heterocycles

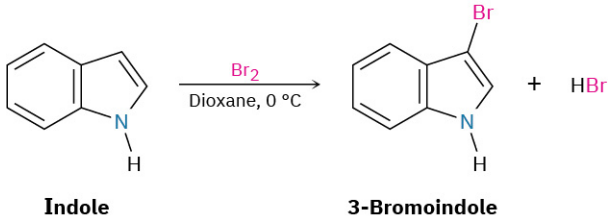
As we saw in **Section 15.6**, quinoline, isoquinoline, indole, and purine are common polycyclic heterocycles. The first three contain both a benzene ring and a heterocyclic aromatic ring, while purine contains two heterocyclic rings joined together. All four ring systems occur commonly in nature, and many compounds with these rings have pronounced physiological activity. The quinoline alkaloid quinine, for instance, is widely used as an antimalarial drug; tryptophan is a common amino acid; and the purine adenine is a constituent of nucleic acids.



The chemistry of these polycyclic heterocycles is just what you might expect from a knowledge of the simpler heterocycles pyridine and pyrrole. Quinoline and isoquinoline both have basic, pyridine-like nitrogen atoms, and both undergo electrophilic substitutions. As with pyridine, both quinoline and isoquinoline are less reactive toward electrophilic substitution than benzene because of the electronegative nitrogen atom that withdraws electrons from the ring. Reaction occurs on the benzene ring rather than on the nitrogen-containing pyridine ring, and a mixture of substitution products is obtained.

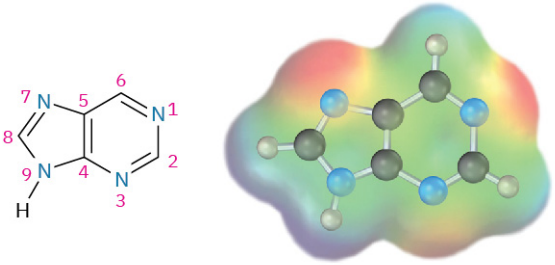


Indole has a nonbasic, pyrrole-like nitrogen and undergoes electrophilic substitution more easily than benzene. Substitution occurs at C3 of the electron-rich pyrrole ring rather than on the benzene ring.



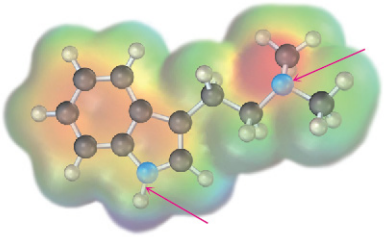
Purine has three basic, pyridine-like nitrogens with lone-pair electrons in sp^2 orbitals in the plane of the ring. The remaining purine nitrogen is nonbasic and pyrrole-like, with its lone-pair electrons as part of the aromatic

π electron system.



Purine

PROBLEM 24-23 Which nitrogen atom in the hallucinogenic indole alkaloid *N*,*N*-dimethyltryptamine is more basic? Explain.



***N*,*N*-Dimethyltryptamine**

PROBLEM 24-24 Indole reacts with electrophiles at C3 rather than at C2. Draw resonance forms of the intermediate cations resulting from reaction at C2 and C3, and explain the observed results.

24.10 Spectroscopy of Amines

Infrared Spectroscopy

Primary and secondary amines can be identified by a characteristic N–H stretching absorption in the 3300 to 3500 cm^{-1} range of the IR spectrum. Alcohols also absorb in this range (**Section 17.11**), but amine absorption bands are generally sharper and less intense than hydroxyl bands. Primary amines show a pair of bands at about 3350 and 3450 cm^{-1} from the symmetric and asymmetric stretching modes, and secondary amines show a single band at 3350 cm^{-1}. Tertiary amines have no absorption in this region because they have no N–H bonds. FIGURE 24.9 recalls the IR spectrum of cyclohexylamine from **Section 12.8**.

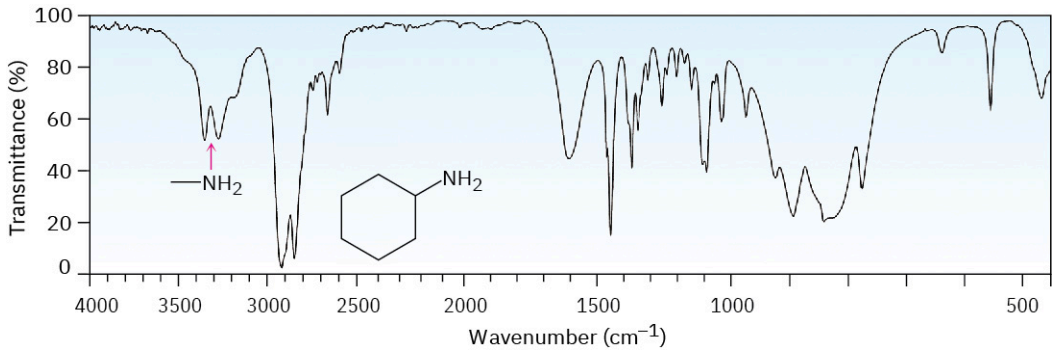
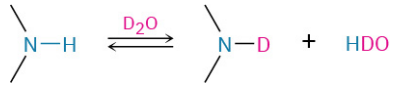


FIGURE 24.9 IR spectrum of cyclohexylamine.

Nuclear Magnetic Resonance Spectroscopy

Amines are difficult to identify solely by ^{1}H NMR spectroscopy because N–H hydrogens tend to appear as broad signals without clear-cut coupling to neighboring C–H hydrogens. As with O–H absorptions (**Section 17.11**), amine N–H absorptions can appear over a wide range and are best identified by adding a small amount of D_2O to the sample. Exchange of N–D for N–H occurs, and the N–H signal disappears from the NMR spectrum.



Hydrogens on the carbon next to nitrogen are deshielded because of the electron-withdrawing effect of the nitrogen, and they therefore absorb further downfield than alkane hydrogens. *N*-Methyl groups are particularly distinctive because they absorb as a sharp three-proton singlet at 2.2 to 2.6 δ. This *N*-methyl resonance at 2.42 δ is easily seen in the ^{1}H NMR spectrum of *N*-methylcyclohexylamine (**FIGURE 24.10**).

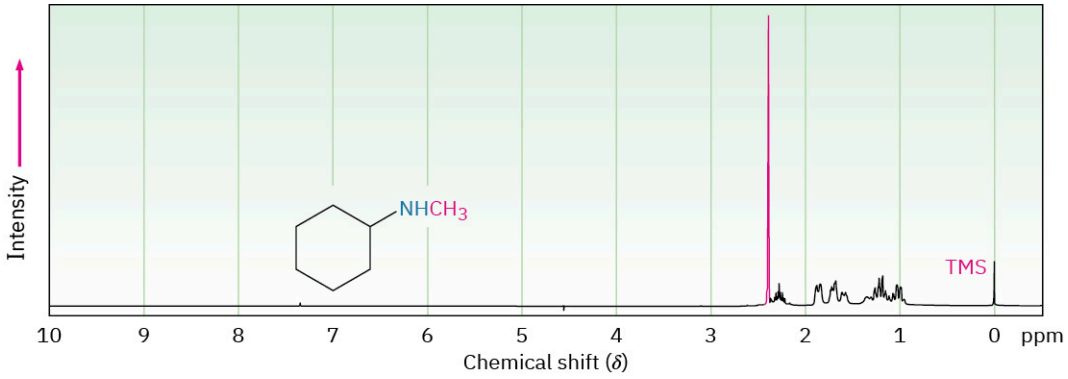
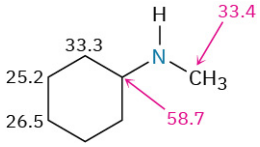
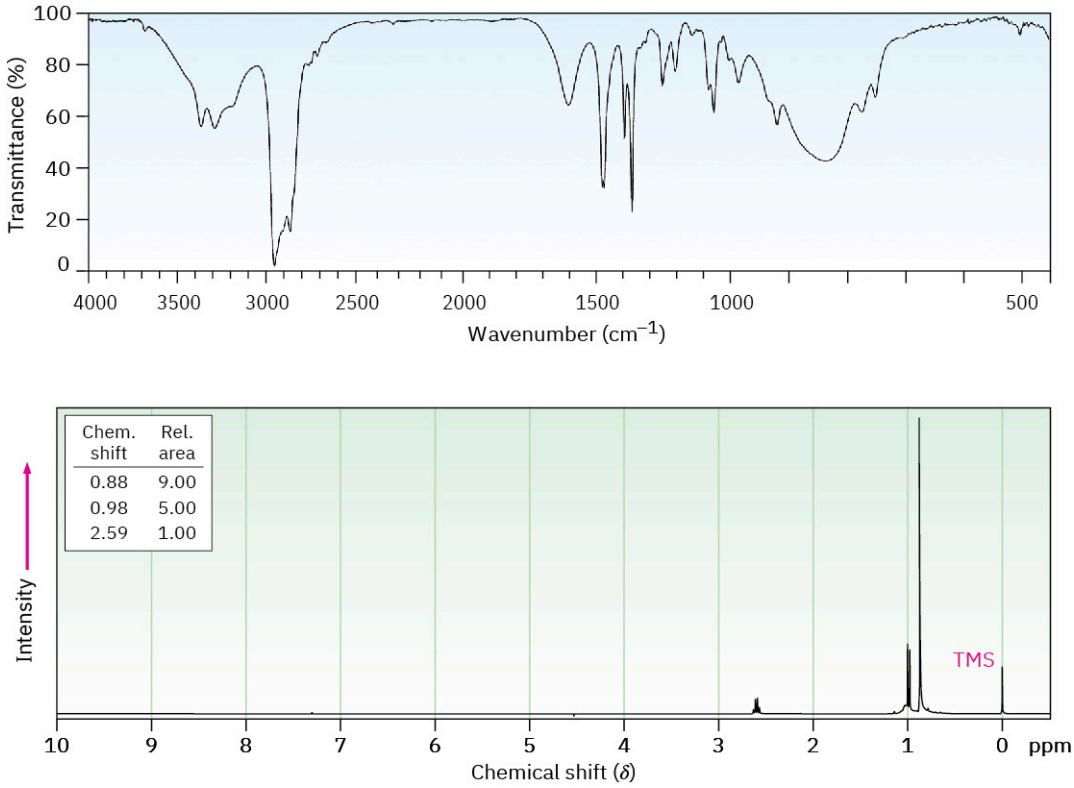


FIGURE 24.10 Proton NMR spectrum of *N*-methylcyclohexylamine.

Carbons next to amine nitrogens are slightly deshielded in the ^{13}C NMR spectrum and absorb about 20 ppm downfield from where they would absorb in an alkane of similar structure. In *N*-methylcyclohexylamine, for example, the ring carbon to which nitrogen is attached absorbs at a position 24 ppm downfield from any other ring carbon.



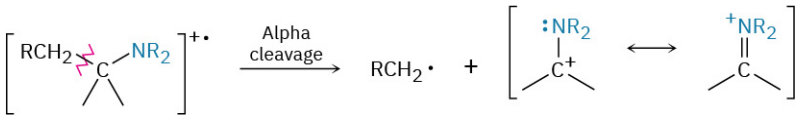
PROBLEM 24-25 Compound **A**, $C_6H_{12}O$, has an IR absorption at 1715 cm^{-1} and gives compound **B**, $C_6H_{15}N$, when treated with ammonia and $NaBH_4$. The IR and ^{1}H NMR spectra of **B** are shown. What are the structures of **A** and **B**?



Mass Spectrometry

The *nitrogen rule* of mass spectrometry says that a compound with an odd number of nitrogen atoms has an odd-numbered molecular weight. Thus, the presence of nitrogen in a molecule is detected simply by observing its mass spectrum. An odd-numbered molecular ion usually means that the unknown compound has one or three nitrogen atoms, and an even-numbered molecular ion usually means that a compound has either zero or two nitrogen atoms. The logic behind the rule derives from the fact that nitrogen is trivalent, thus requiring an odd number of hydrogen atoms. For example, morphine has the formula $C_{17}H_{19}NO_3$ and a molecular weight of 285 amu.

Alkylamines undergo a characteristic α cleavage in the mass spectrometer, similar to the cleavage observed for alcohols (**Section 17.11**). C–C bond nearest the nitrogen atom is broken, yielding an alkyl radical and a resonance-stabilized, nitrogen-containing cation.



As an example, the mass spectrum of *N*-ethylpropylamine shown in **FIGURE 24.11** has peaks at $m/z = 58$ and $m/z = 72$, corresponding to the two possible modes of α cleavage.

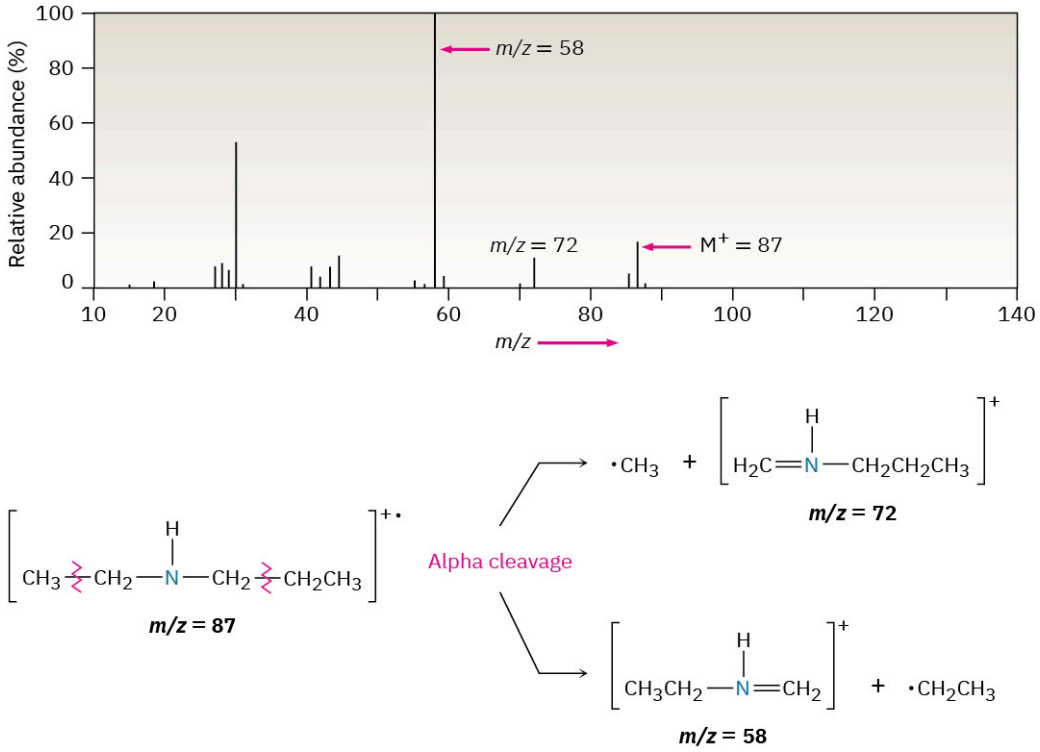


FIGURE 24.11 Mass spectrum of *N*-ethylpropylamine. The two possible modes of α cleavage lead to the observed fragment ions at m/z = 58 and m/z = 72.